

Quantitative model

Verified history and model fit

13 September 2026, 13:48 UTC | Six-period history and paired tax policies verified; horizon adequacy pending.

The fiscal and distribution repairs pass. The economic fit remains weak. The corrected initial equilibrium reproduces exactly in two independent numerical repetitions. Its objective is 179.298424. This is a verified candidate from an incomplete search, not a converged calibration optimizer.

Question	Verified evidence
Does the pension budget balance?	Yes in the initial equilibrium and every accepted historical forecast date.
Is property tax rebated?	Yes. Revenue returns equally per current household head; the rebate budget is checked separately from pensions.
Was the probability error fixed?	Yes. Float64 normalization in both Markov distribution operators passes the captured mass-error replay and full initial reconstruction.
Is the entire shock path fitted?	Yes for the provisional six-period A0 history. All four observed fertility windows pass the 0.005 fit requirement. Horizon adequacy remains unverified.
Are policy results ready?	Yes for the six-period A0 case: baseline and doubled property tax both return revenue equally, balance pensions, and start from identical 2023 households.

What deserves attention

Mean rooms and ownership at ages 30-55 are the largest remaining fit problems. Annual beta reaches its 0.99 cap. The numerical repair changes the objective by less than one millionth; it is not evidence of improved economic fit.

The 6-, 24- and 100-period forecasts are separate horizon checks. Passing a finite market-clearing root does not establish that a shorter horizon is adequate. The alternative demographic closure remains unresolved and has not been promoted.

The paired tax exercise changes the first birth flow by -0.25% and the final simulated birth flow by +0.47%. These are small, provisional effects. A0 assumes zero net migration after 2023; six periods cover 24 years, followed by a constant-condition boundary for remaining household lifetimes.

How the historical exercise works

At each surprise, households expect the new fertility preference to remain constant. For guessed paths of house prices, pensions and rebates, the solver works backward over household choices and forward over inherited households. It adjusts those paths until markets and both fiscal budgets clear, then adjusts the preference to match that observation window. Only the first realized period is carried to the next surprise.

Complete initial target fit

All 13 retained rows: 12 scored moments and the separate completed-fertility normalization. Objective 179.298424. The target system and weights are unchanged.

Moment	Target	Model	Gap	Weight	Loss
Initial model completed fertility	2.1	2.1	8.22549e-07	-	-
Childless women, ages 40–44	0.198279	0.196258	-0.00202037	35532.3	0.145039
Exactly one child among mothers, ages 40–44	0.213655	0.216439	0.00278364	26952.8	0.208848
Period mean first-birth age	25.9763	26.0952	0.118982	139.828	1.9795
First births at age 30+	0.249278	0.232155	-0.0171228	13866.1	4.06539
Wealth / annual gross labor earnings	6.14586	4.89343	-1.25243	7.5951	11.9136
Annual bequests / aggregate wealth	0.0088	0.00858241	-0.000217591	5.16529e+06	0.244554
Old wealth/income p90 / median, ages 76–84	3.51594	4.4998	0.983865	10.6164	10.2765
Mean occupied rooms, capped at 9	5.5611	6.42465	0.863555	128.021	95.4685
Ownership, heads 30–55	0.648334	0.539896	-0.108438	2339.36	27.508
First-birth room response, -1 to +3	0.720246	0.980806	0.26056	137.565	9.3395
Rooms: 3+ versus 1–2 resident children (model dependent proxy)	0.347067	0.172043	-0.175024	280.528	8.59354
Recent-parent ownership gap	0.162896	0.144103	-0.0187928	27055.8	9.55533

The normalization row does not enter the loss. Units follow each empirical moment: ownership is a fraction, room responses are in rooms, and first-birth age is in years. The CSV preserves every row's builder, sample, vintage, observation definition, uncertainty and weighting provenance.

Interpretation

Initial fertility levels are close by construction and the fertility-composition moments fit relatively well. Mean rooms are too high and ownership is too low. The first-birth room response overshoots its target. A numerically valid equilibrium therefore does not imply an adequate fit across the housing mechanisms.

Historical observation clock

Decision vintage	Observed TFR window	Target
2007	2008-2011	1.974875
2011	2012-2015	1.861000
2015	2016-2019	1.755375
2019	2020-2023	1.645750

A 2023 decision-vintage fertility flow belongs to the 2024-2027 forecast. It is not an additional fitted historical observation. Other 2023 model/data comparisons must use the correctly dated inherited economy.

Verified four-window history

All four fertility windows are matched in a six-period forecast. Each accepted date passes housing, PAYGO and equal-rebate gates. Horizon adequacy remains unverified; longer forecasts continue. Matching the fertility decline does not resolve weak cross-sectional fit.

Window end	Data	Model	Gap	psi
2011	1.97487	1.97312	-0.00175075	0.128915
2015	1.861	1.86104	4.20102e-05	0.116966
2019	1.75537	1.75534	-3.68441e-05	0.105643
2023	1.64575	1.64556	-0.000190682	0.0922185

Full 2023 validation: Data / Model (untargeted)

These comparisons are a validation readout; they are not additional fitted targets.

Moment	Data	Model	Gap	Vintage
Completed fertility, ages 40–44	1.91842	1.69387	-0.224557	CPS 2024 (nearest fertility supplement)
Childless, ages 40–44 (%)	18.818	24.0122	5.19426	CPS 2024 (nearest fertility supplement)
Exactly one child among mothers, 40–44 (%)	22.2289	26.3297	4.10072	CPS 2024 (nearest fertility supplement)
Mean age at first birth (years)	28.0902	29.7975	1.70727	NCHS 2023
First births at age 30+ (%)	37.6557	44.0911	6.43539	NCHS 2023
Mean occupied rooms (capped at 9)	5.58372	6.25308	0.669353	ACS 2023
Ownership, heads 30–55 (%)	58.7409	49.4514	-9.28948	ACS 2023
First-birth room response, –1 to +3	0.720246	1.01368	0.293429	PSID pooled; retained Sun–Abraham receipt
Rooms: 3+ versus 1–2 resident children	0.354798	0.222455	-0.132343	ACS 2023
Recent-parent ownership gap (pp)	21.1161	13.756	-7.3601	ACS 2023
Wealth / annual gross earnings	6.87308	5.61438	-1.25869	PSID 2005–2019
Annual bequests / wealth (%)	0.88	0.714865	-0.165135	External benchmark
Wealth/income p90 / median, ages 76–84	3.44811	3.49523	0.0471162	PSID 1984–2019

2023 validation measurement caveats

Completed fertility, ages 40–44. Children ever born stock; distinct from period fertility.

Mean age at first birth (years). Model 2024–2027 flow versus annual 2023 NCHS births.

First births at age 30+ (%). Model 2024–2027 flow versus annual 2023 NCHS births.

First-birth room response, –1 to +3. Sun–Abraham empirical contrast; model matched branch from 2019 into 2023.

Rooms: 3+ versus 1–2 resident children. Model dependent counts proxy resident own children under 18.

Recent-parent ownership gap (pp). Model current birth into an empty dependent home versus currently empty home; retained flow proxy, not exact ACS oldest-child-age reconstruction.

Annual bequests / wealth (%). External historical restriction; not a 2023 observation.

Wealth/income p90 / median, ages 76–84. Beginning-period wealth of living households; model pension-income proxy versus empirical family income.

Policy comparison: 2023–2043

A0 means no net migration after 2023. The six periods span 24 years, with a remaining-lifetime constant-condition boundary. The paired reforms apply a 1% annual property tax versus a 2% annual property tax, with equal rebates. Both start from the same inherited 2023 households and balance PAYGO separately. Births are per-period flows, not cumulative; the last decision in 2043 corresponds to the 2044–47 flow. These are provisional short-horizon results because horizon adequacy is unverified.

Decision year 2023

Metric	Baseline	2% tax	Percent change	Change (pp)
Asset price	0.813077	0.759016	-6.64892%	
Renter price	0.130606	0.15699	20.2015%	
Housing demand	7.27933	6.97093	-4.23673%	
Resident persons	2.96598	2.96598	0%	
Household heads	1.12925	1.12925	3.16575e-12%	
Births (flow)	0.107858	0.107591	-0.247838%	
Owner rate	0.570734	0.555867	-2.60491%	-1.48671 pp
Pension (period units)	2.33	2.33	-6.13721e-12%	
Equal transfer (period units)	0.20965	0.374837	78.7921%	
Period TFR (flow)	1.63722	1.6336	-0.221157%	
Rooms per head	6.44618	6.17307	-4.23673%	

Decision year 2043

Metric	Baseline	2% tax	Percent change	Change (pp)
Asset price	0.804484	0.742941	-7.64997%	
Renter price	0.137605	0.15713	14.1891%	
Housing demand	7.23103	6.87751	-4.8889%	
Resident persons	2.8944	2.89534	0.0325142%	
Household heads	1.17832	1.17831	-0.00103055%	
Births (flow)	0.101432	0.10191	0.471848%	
Owner rate	0.584273	0.598536	2.44111%	1.42627 pp
Pension (period units)	1.89784	1.89782	-0.000991987%	
Equal transfer (period units)	0.197476	0.34691	75.6721%	
Period TFR (flow)	1.6375	1.64769	0.622276%	
Rooms per head	6.13673	5.83677	-4.88792%	

Housing supply uses the same price-dependent curve in both policies, with elasticity 0.63. The lower house price reduces supply, so the tax experiment includes a contraction in total housing as well as changes in its allocation. Reported rooms per head are physical rooms; the empirical validation table uses its separate top-coded room measure.

Parameters and restrictions

Nine structural search coordinates. The other rows are fixed restrictions, normalizations or endogenous fiscal outcomes; they are not additional structural estimates.

Parameter	Value	Lower	Upper	Near bound	Search
beta_annual	0.99	0.94	0.99	Yes	Yes
kappa_fert	0.337734	0.02	50	Yes	Yes
kappa_fert_continuation	0.397856	0.02	50	Yes	Yes
chi	1.04965	0.1	5	No	Yes
H0	8.1121	0.2	80	No	Yes
theta0	0.081051	0	8	No	Yes
theta1	0.0852036	0.02	16	Yes	Yes
first_birth_fixed_cost	0.265765	0	8	No	Yes
h_P	2.3	0.1	2.3	Yes	Yes
hbar_child_rooms	0	-	-	No	No
psi_child	0.148915	-	-	No	No
payroll_tax	0.179	-	-	No	No
pension_period	2.04636	-	-	No	No
housing_supply_elasticity	0.63	-	-	No	No
tenure_choice_kappa	0.005	-	-	No	No
alpha_cons	0.733	-	-	No	No
sigma	2	-	-	No	No

The annual-beta upper bound is the enforced 0.99. The scorer's older metadata is preserved in parameters_raw.csv. Near-bound flags use the retained reporting rule and should not be read as proof of binding first-order conditions.

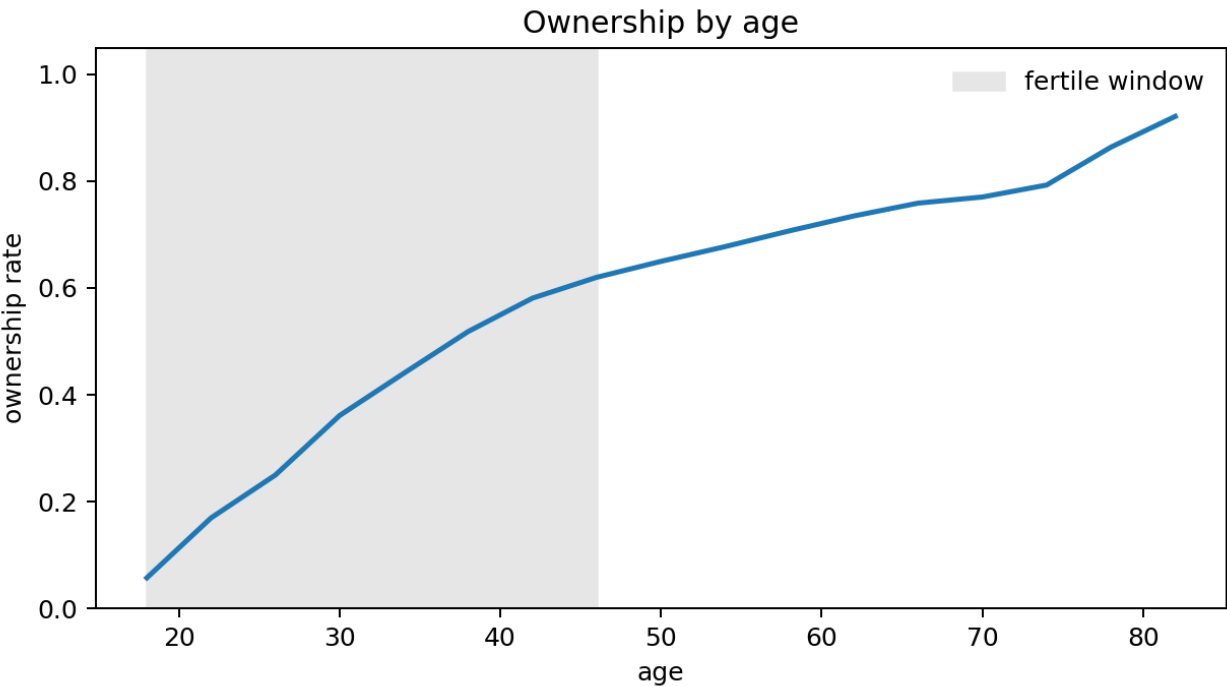
Numerical acceptance evidence

Initial check	Observed value
Stationary nesting, L1	2.2747454e-14
One-period nesting, L1	2.6512812e-13
PAYGO relative budget gap	3.0689413e-12
Equal-rebate relative budget gap	3.8384145e-7
Second numerical signature	Exactly equal

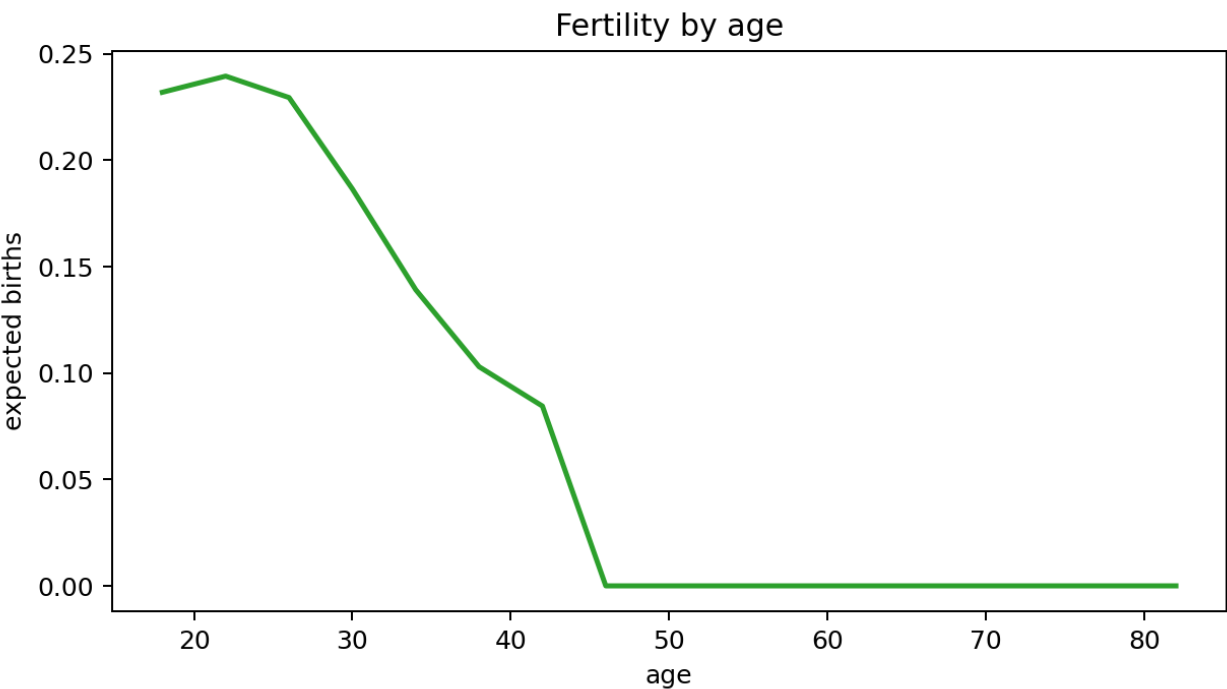
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Native initial diagnostics

The unchanged 17-figure packet from the verified initial solution. These are initial model diagnostics, not fitted 2023 comparisons. Native graph statistics can use different caps or age weights from the calibration observers.



ownership by age



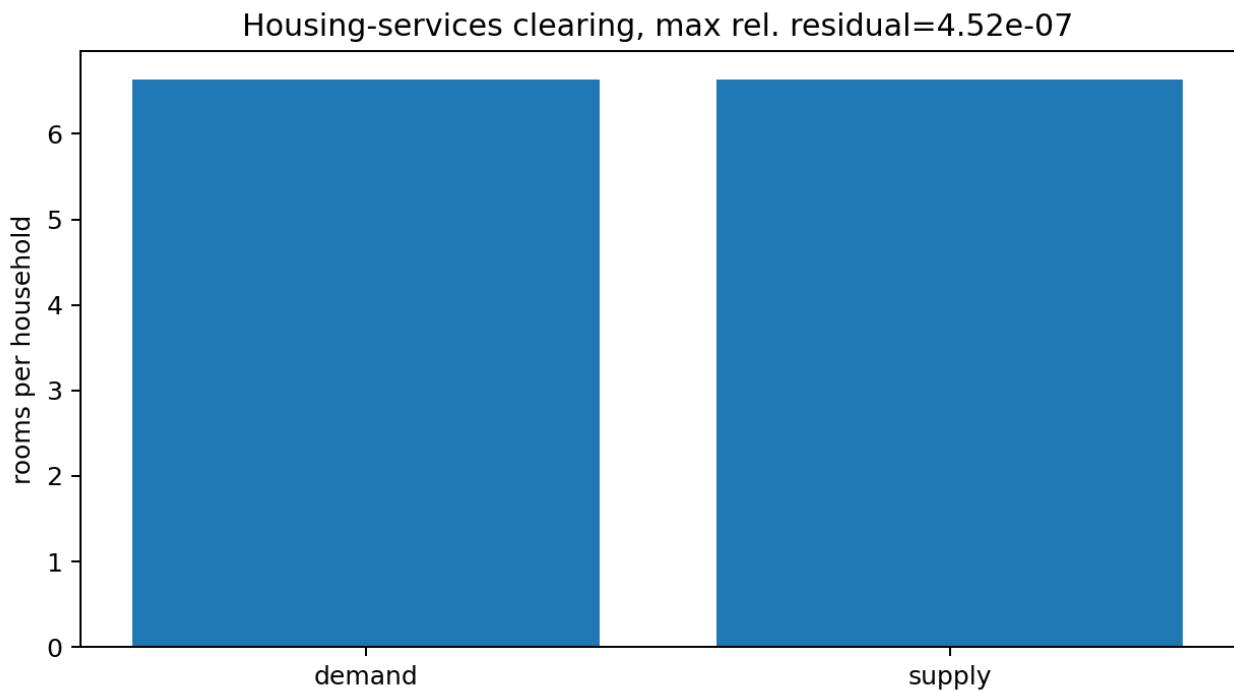
fertility by age

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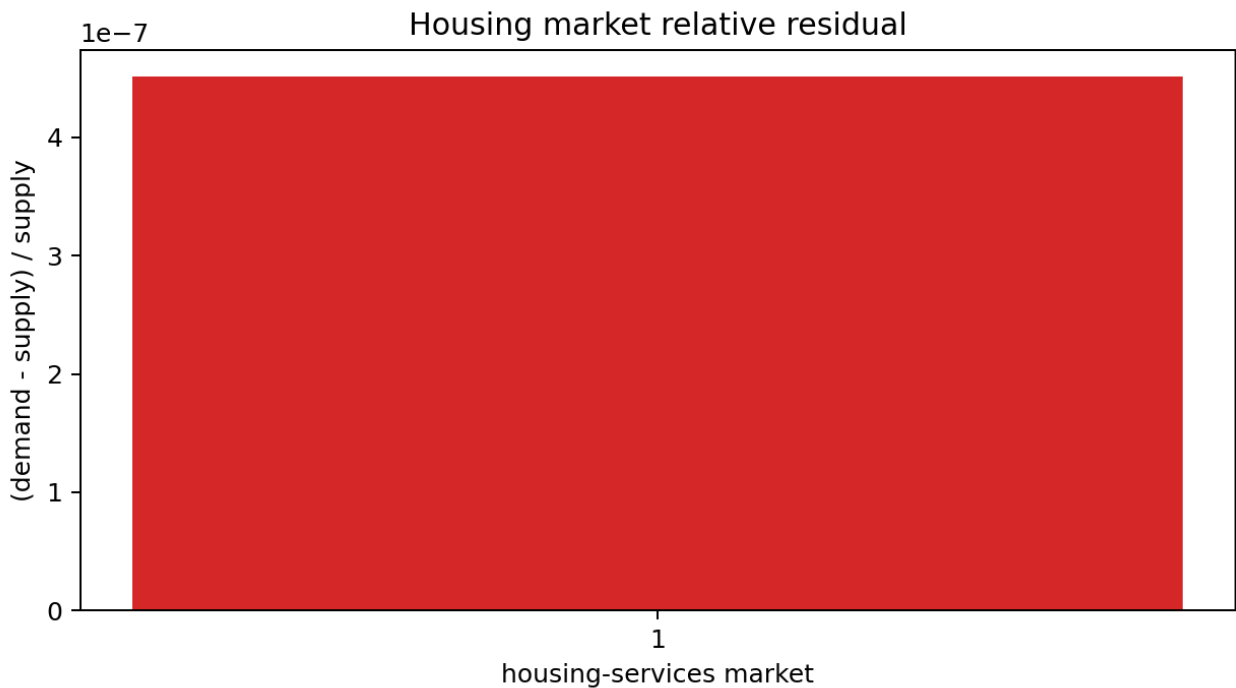
housing prices



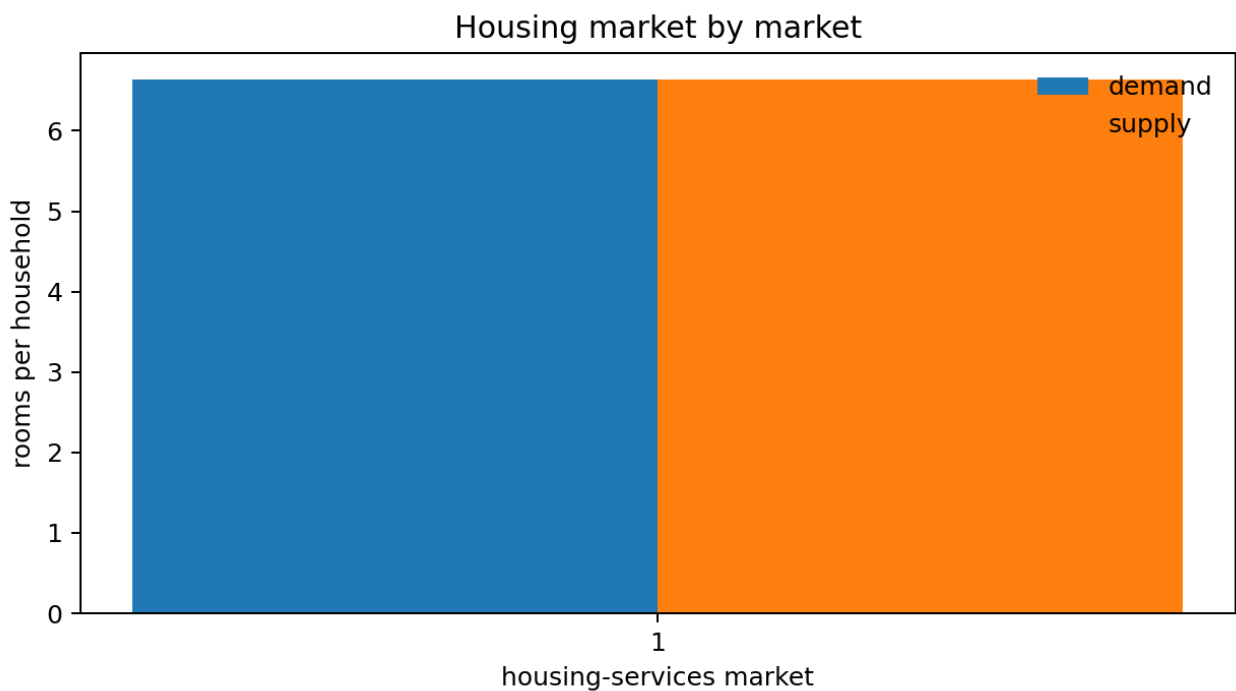
housing market

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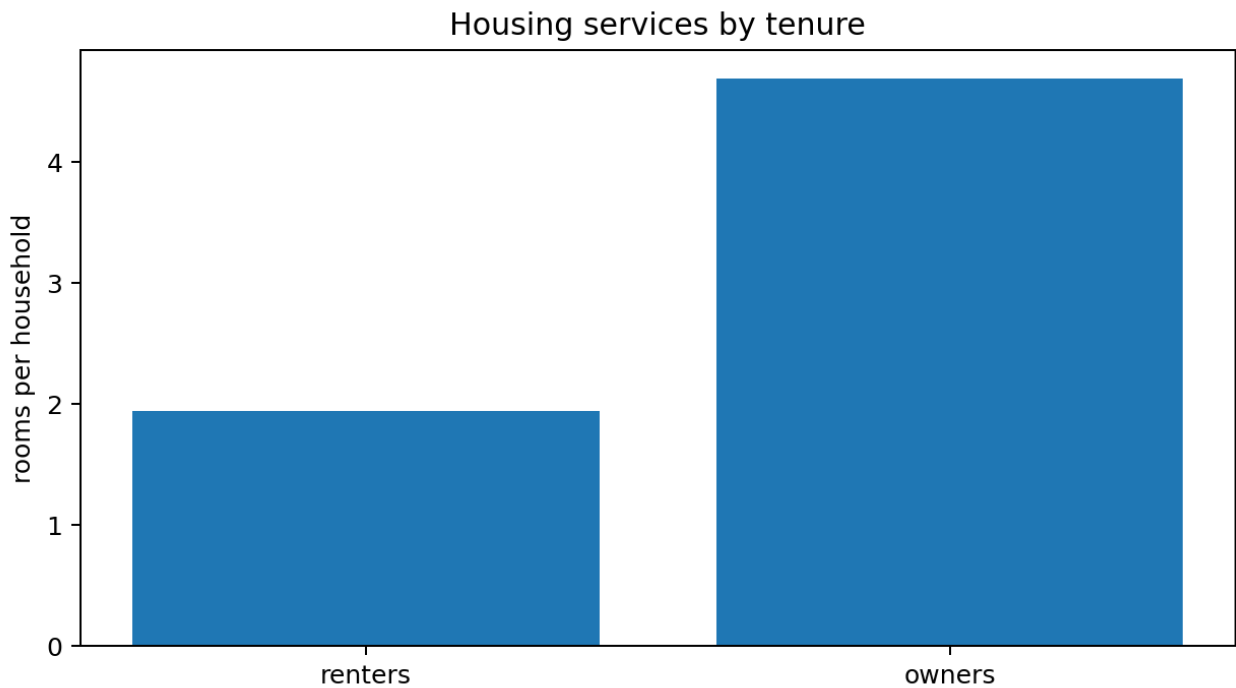
market clearing residuals



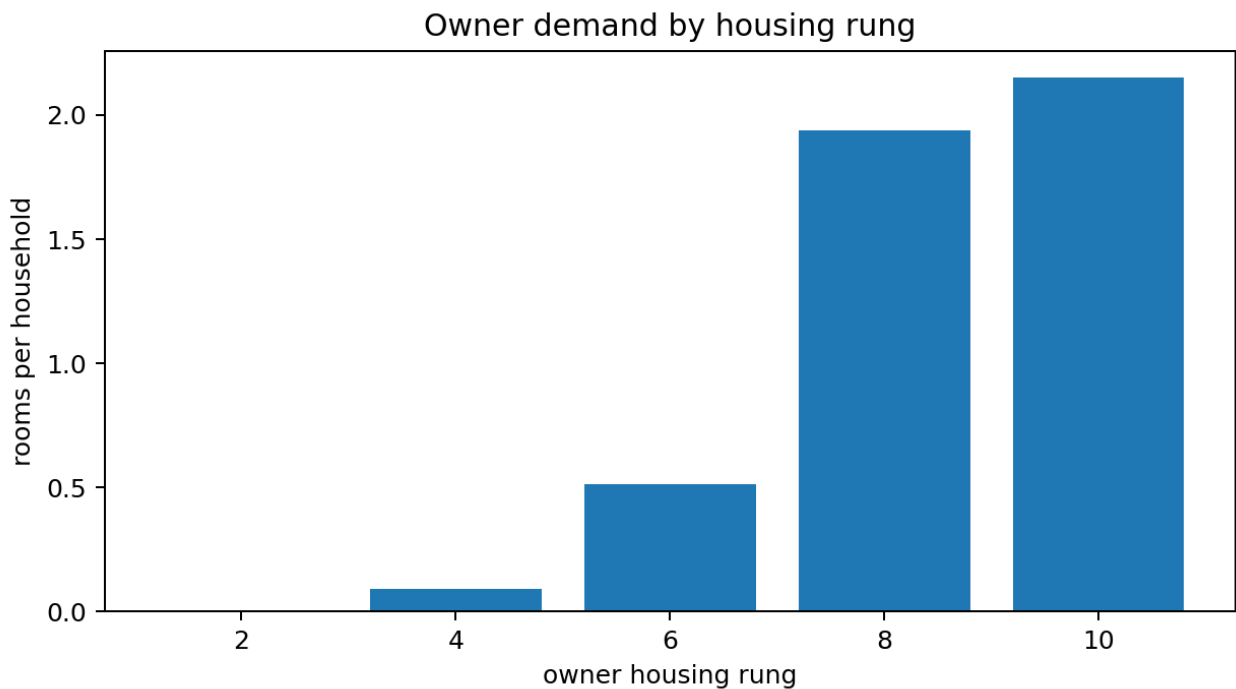
market clearing by market

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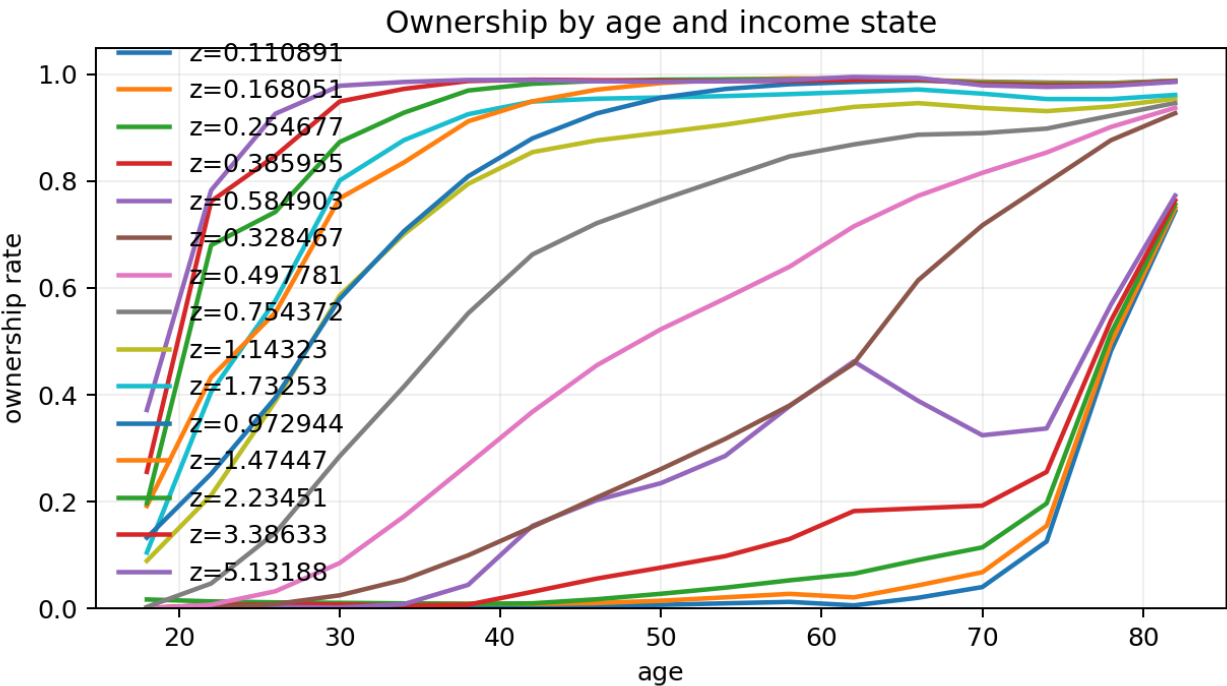
tenure services



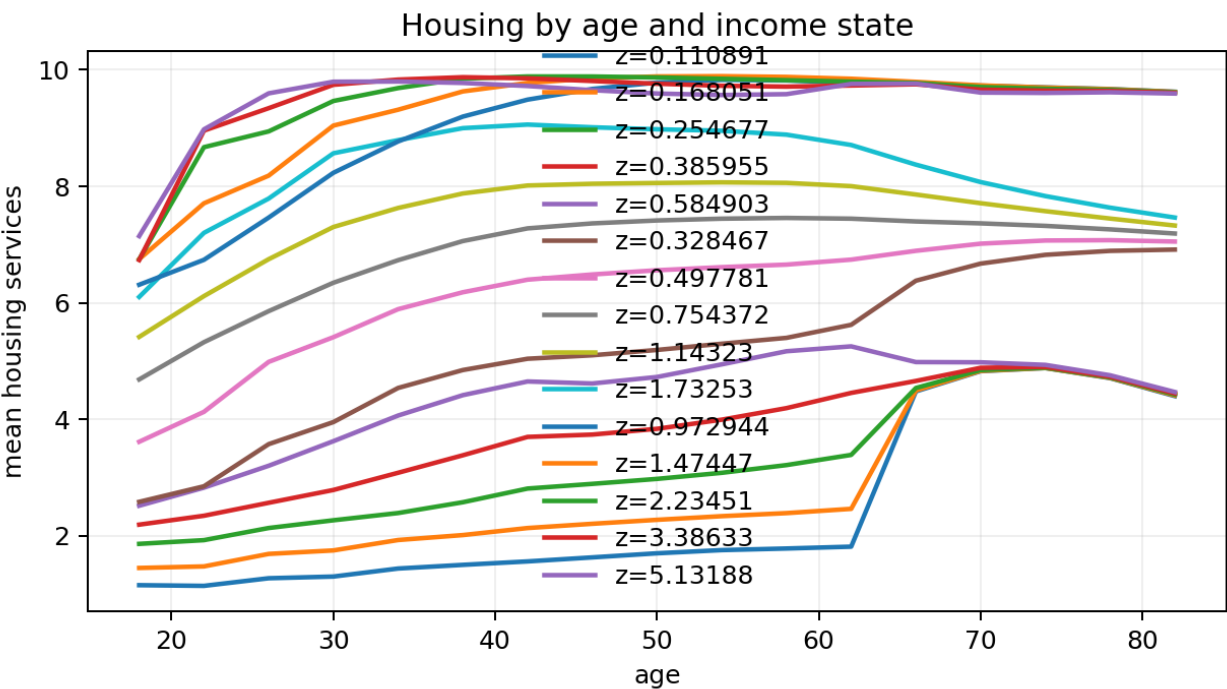
owner rungs

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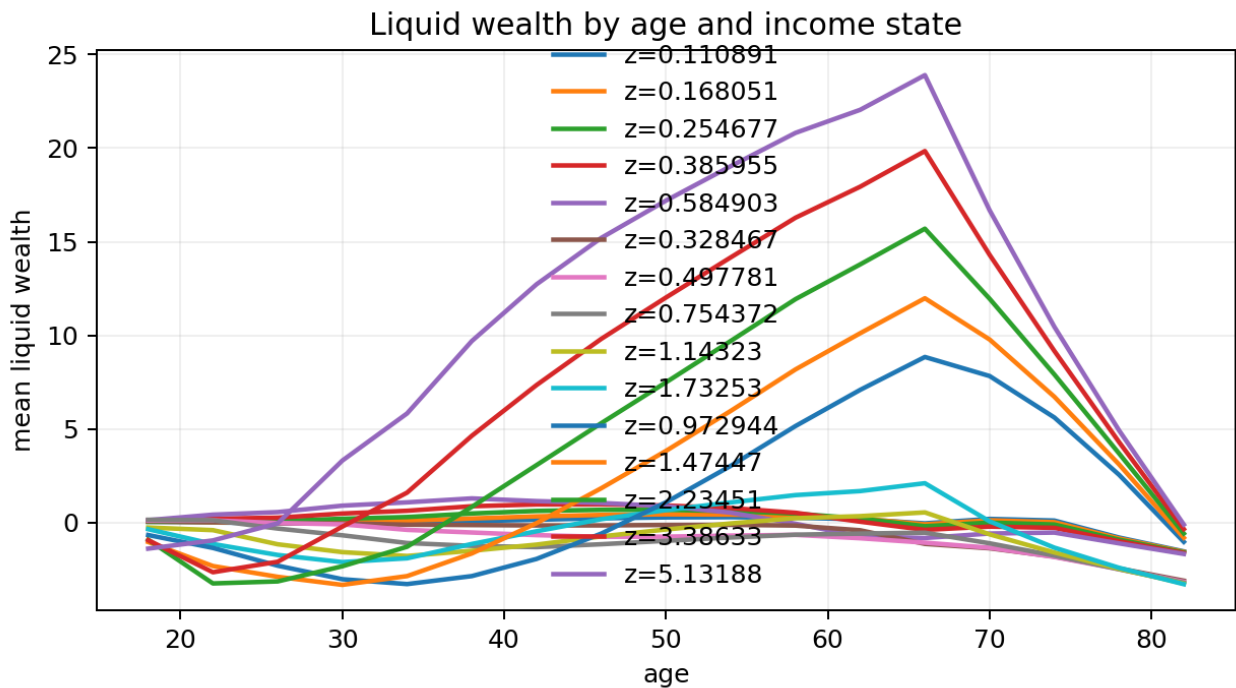
ownership by age income state



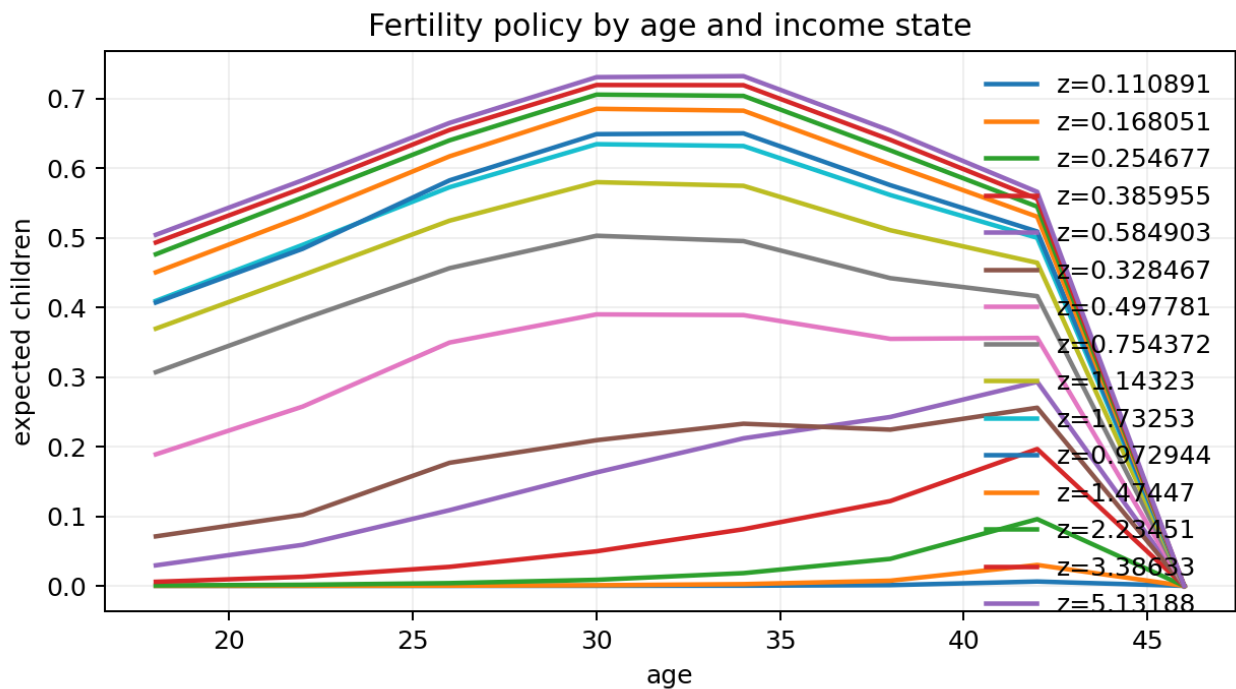
housing by age income state

Native initial diagnostics

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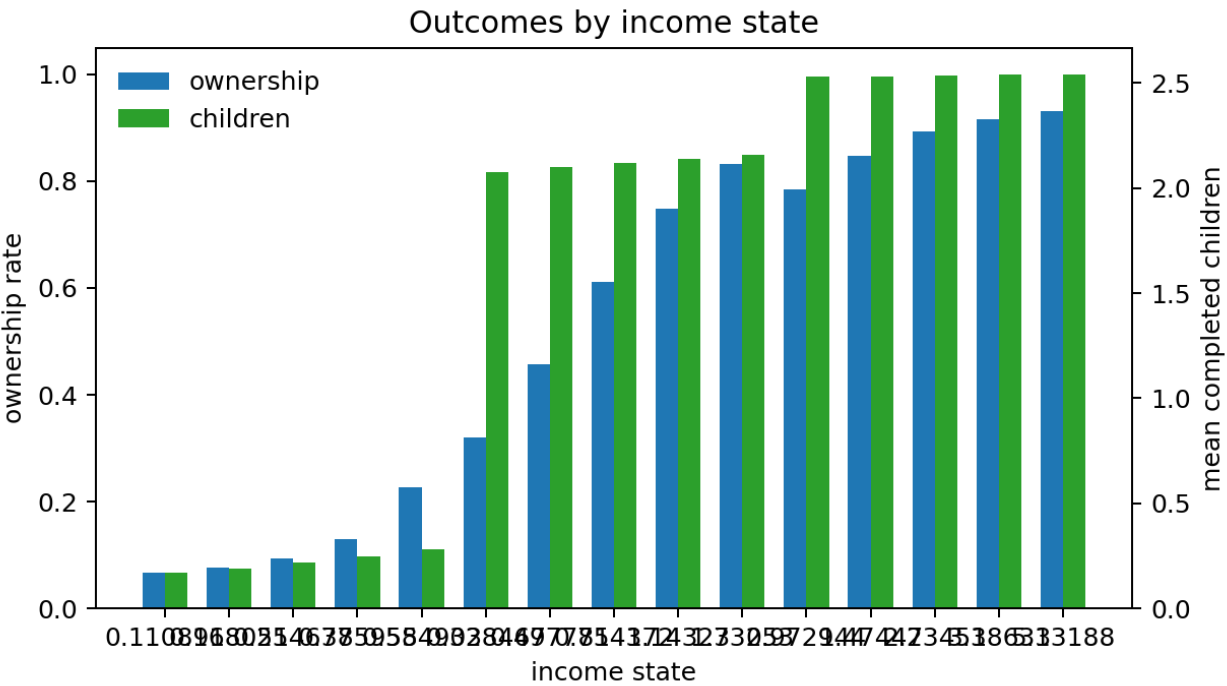
liquid wealth by age income state



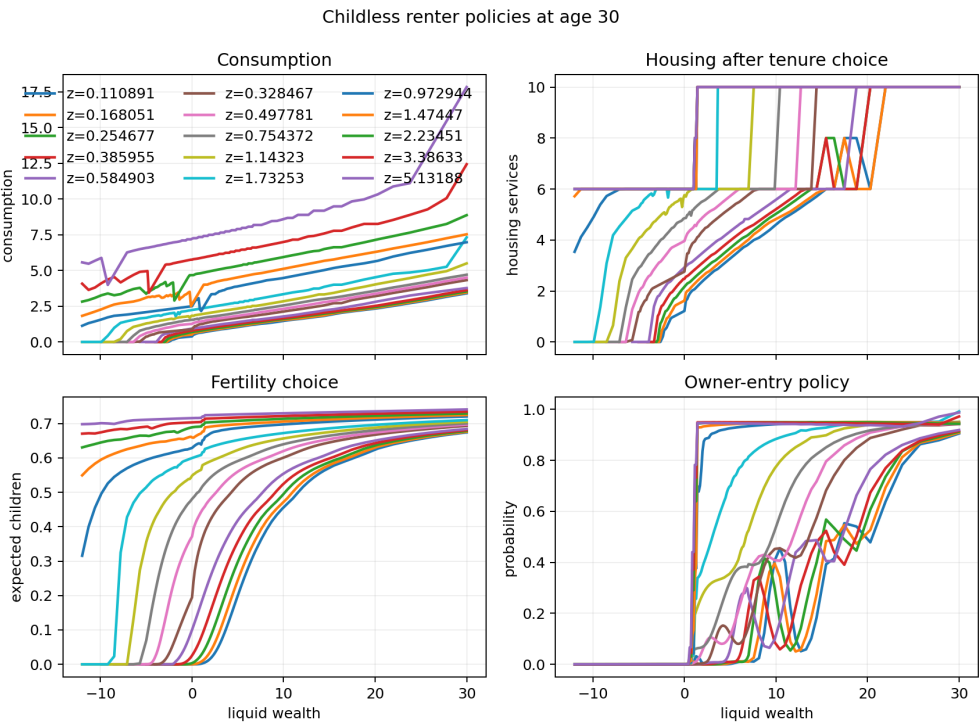
fertility policy by age income state

Native initial diagnostics

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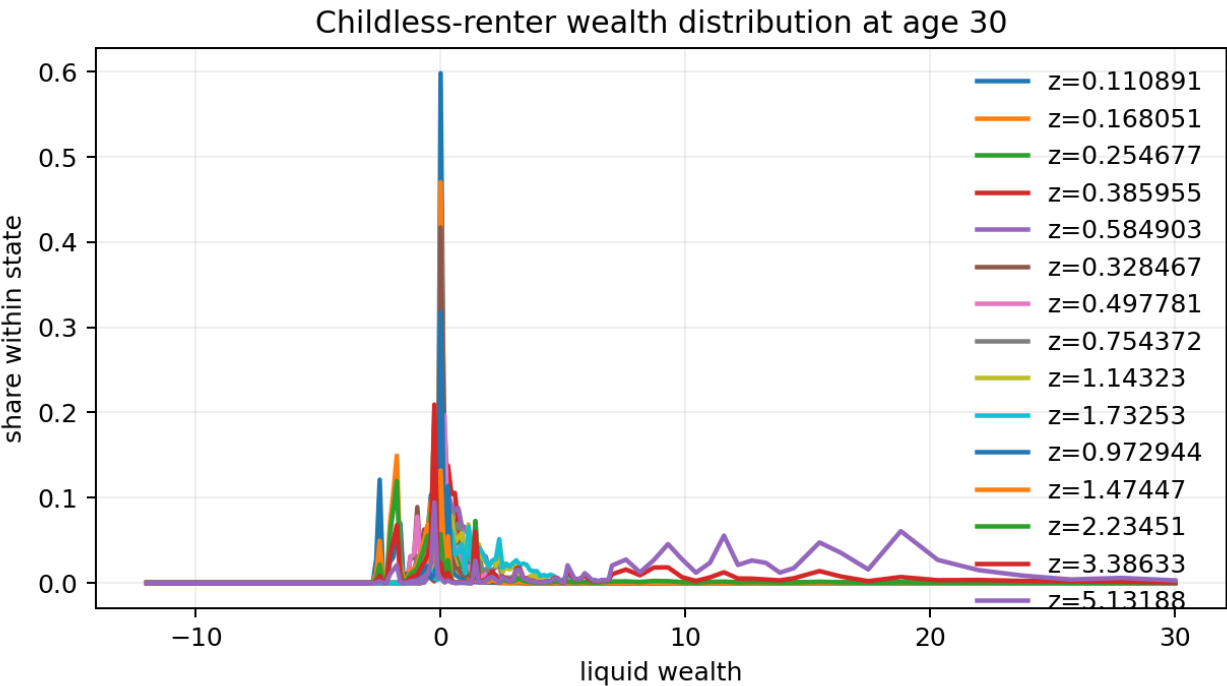
income state outcomes



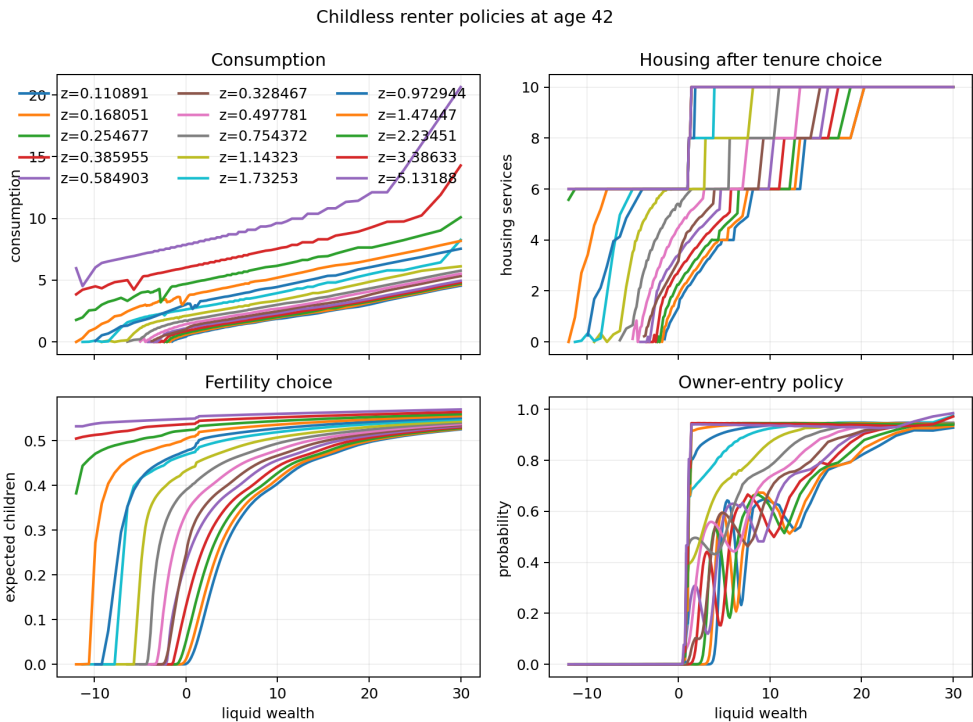
policy childless renter age30

Native initial diagnostics

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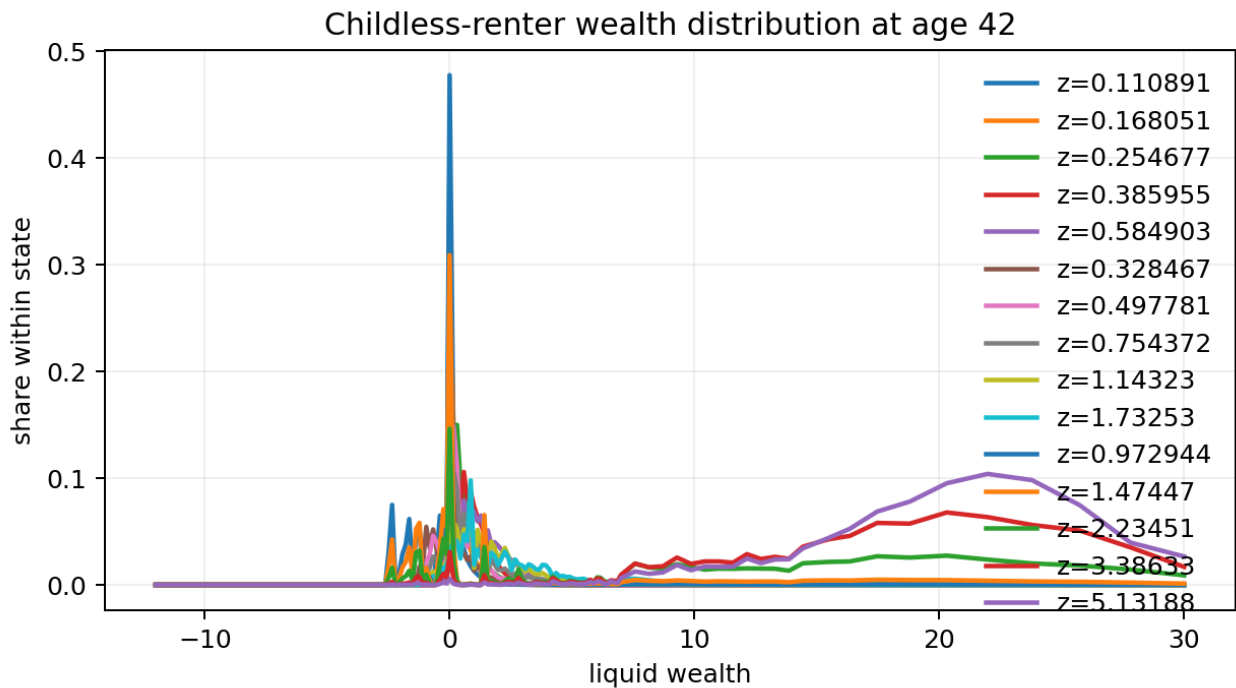
wealth dist childless renter age30



policy childless renter age42

Native initial diagnostics

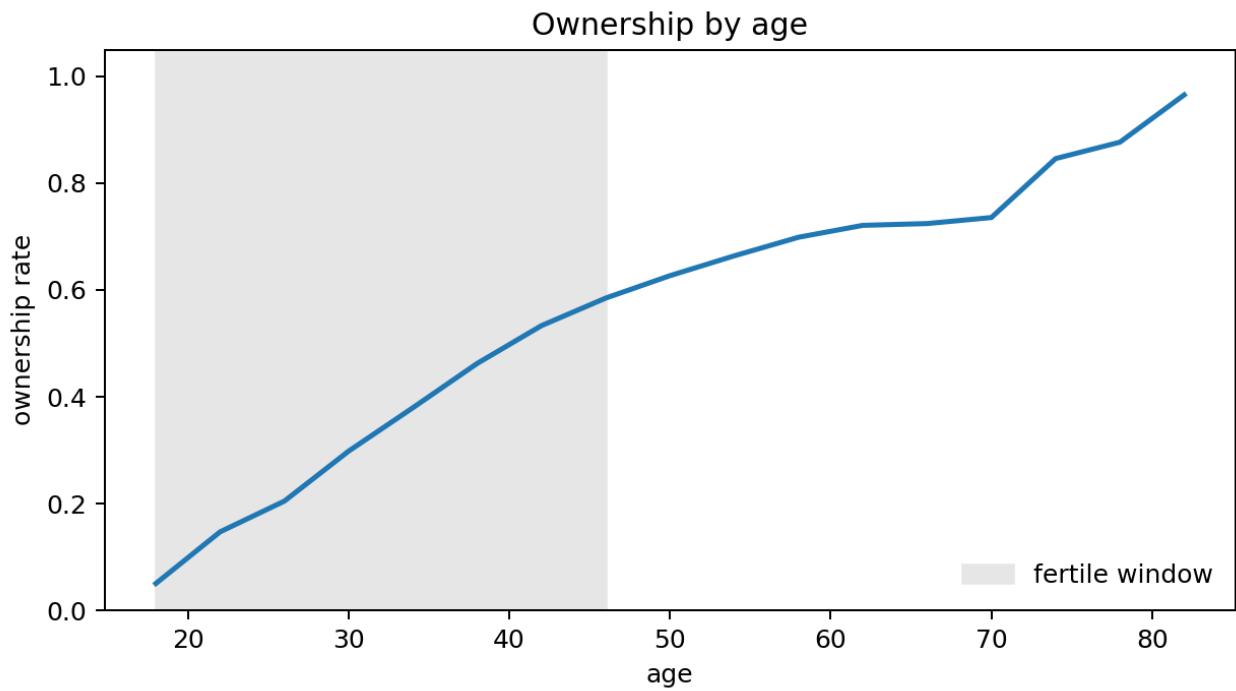
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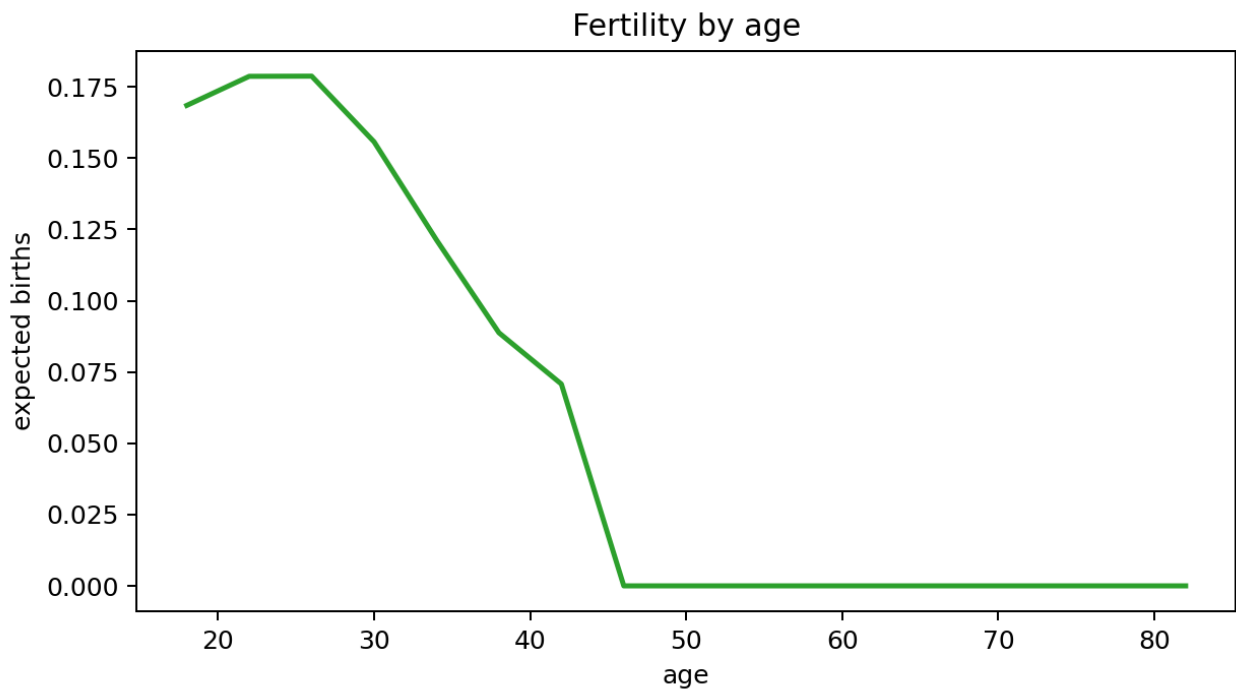
wealth dist childless renter age42

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



Baseline policy — ownership by age



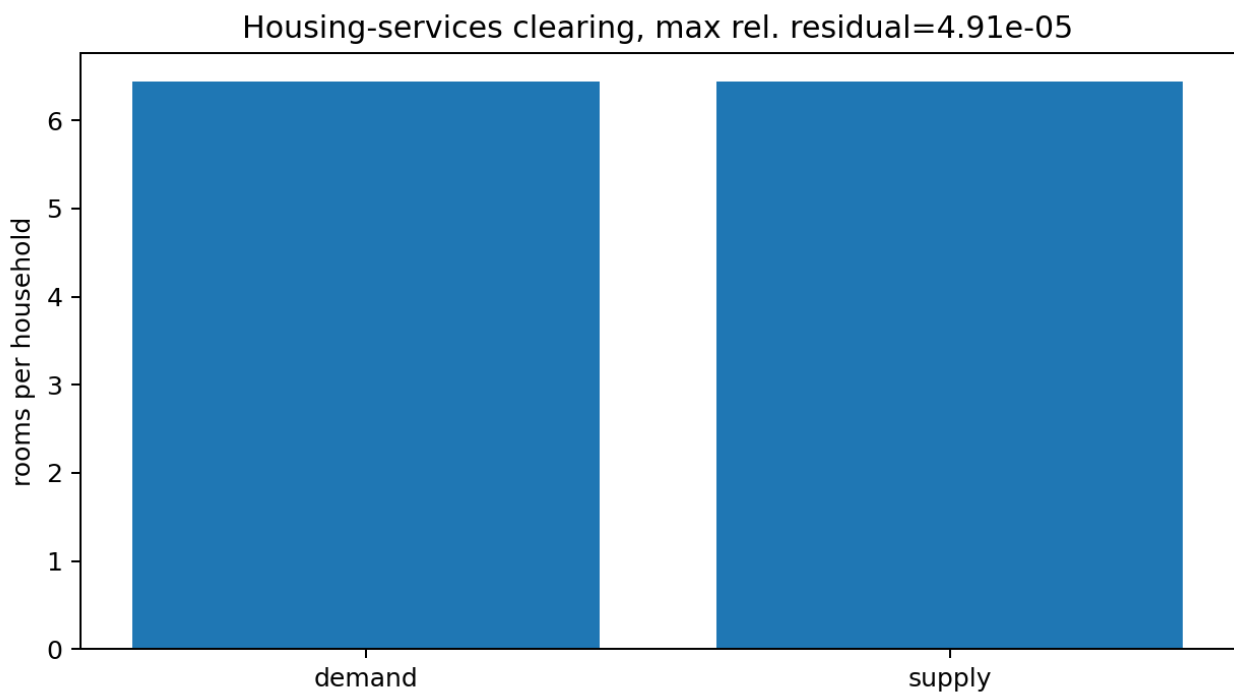
Baseline policy — fertility by age

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



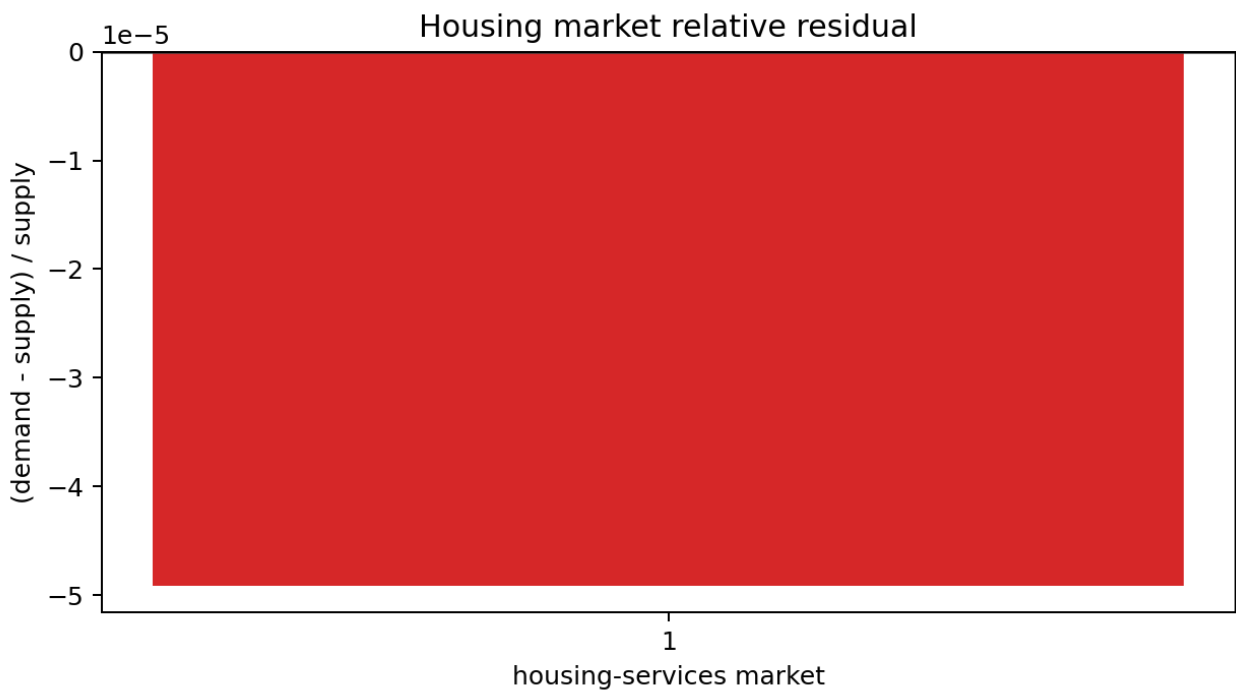
Baseline policy — housing prices



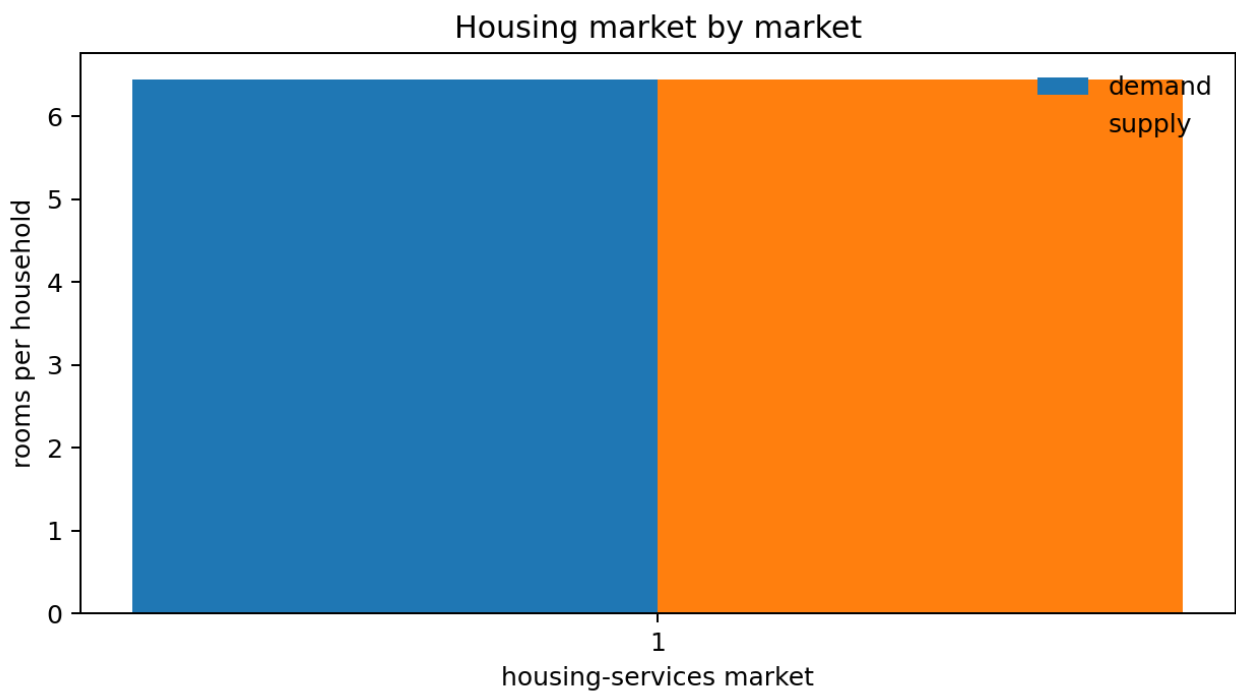
Baseline policy — housing market

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



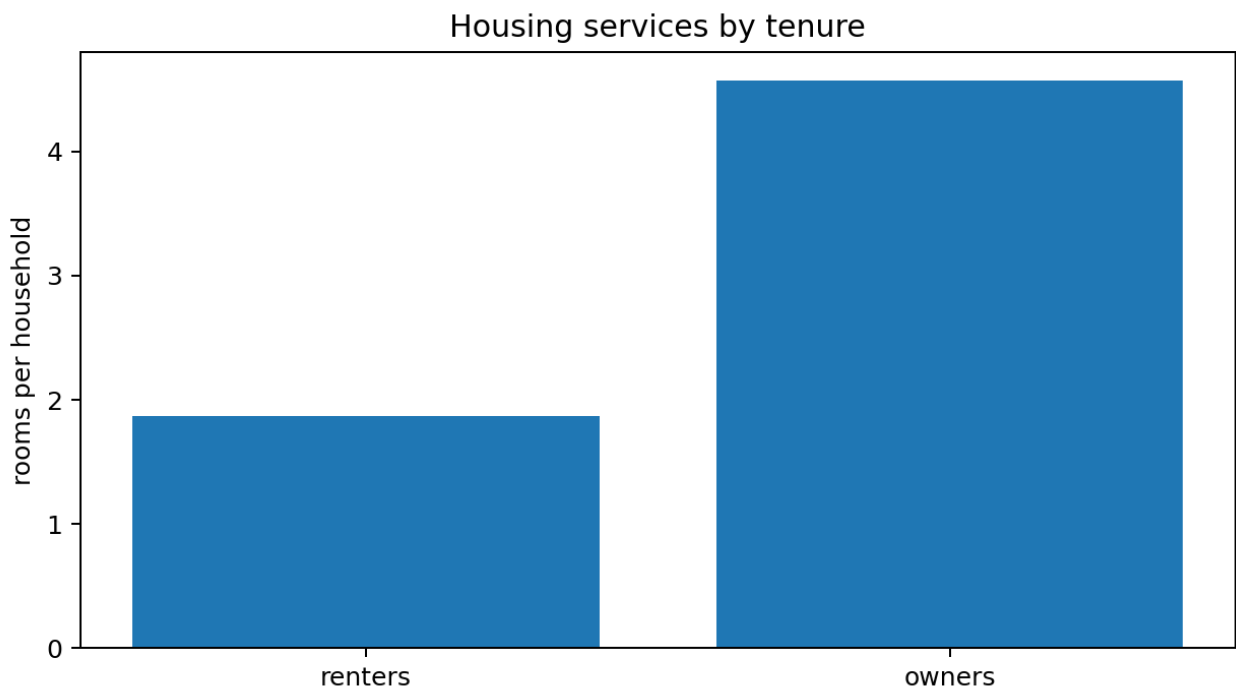
Baseline policy — market clearing residuals



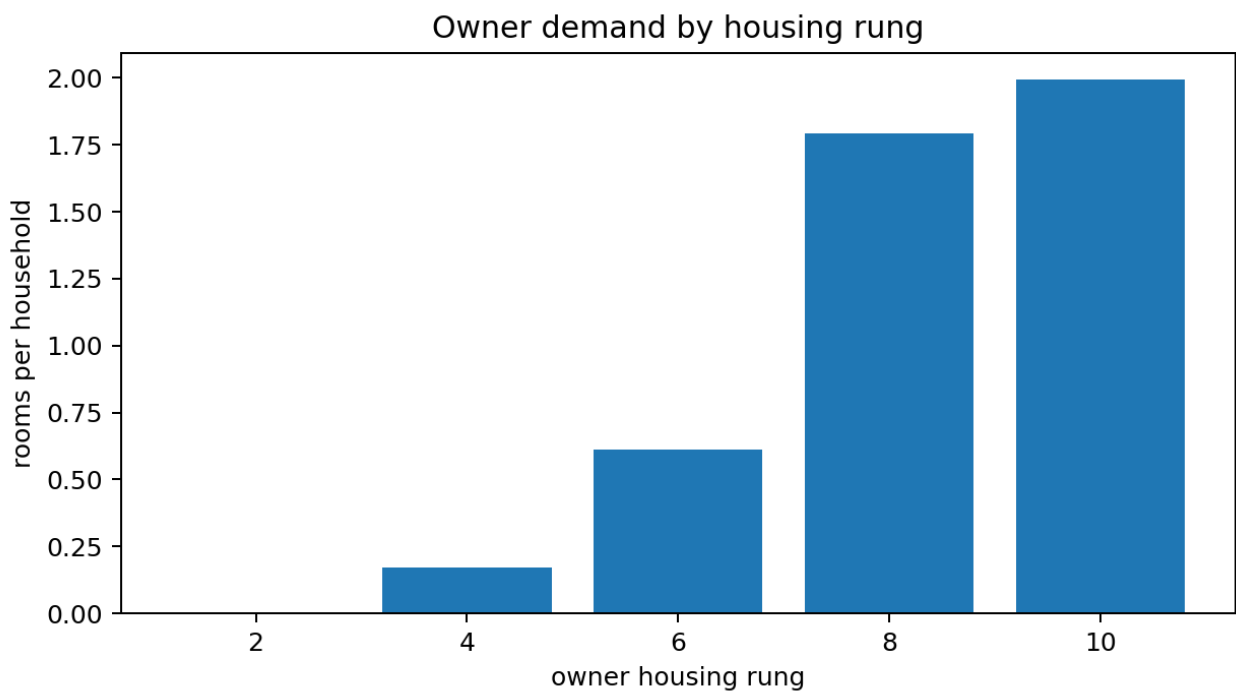
Baseline policy — market clearing by market

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



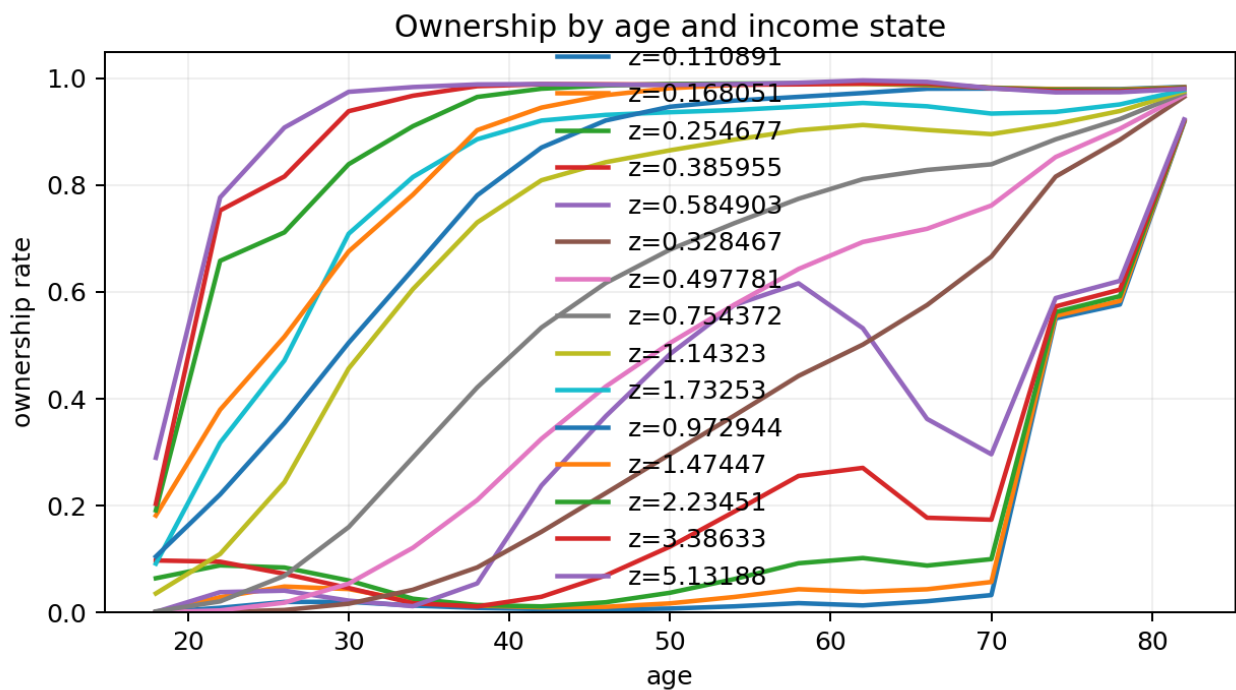
Baseline policy — tenure services



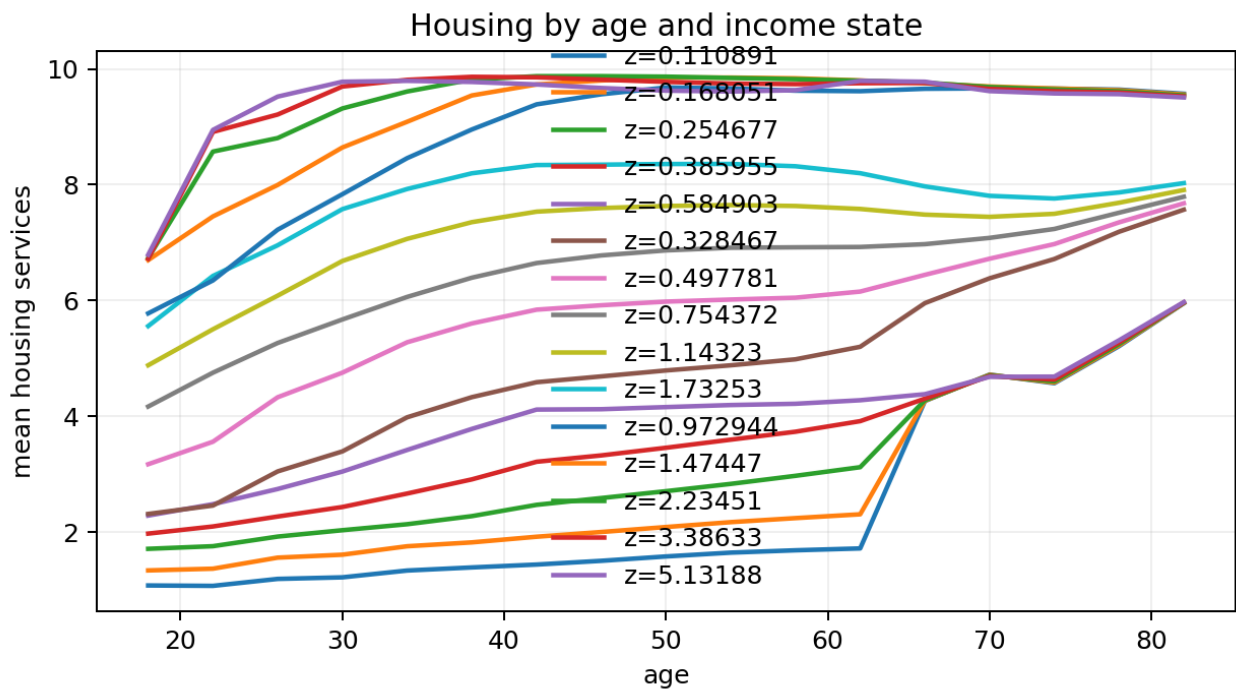
Baseline policy — owner rungs

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



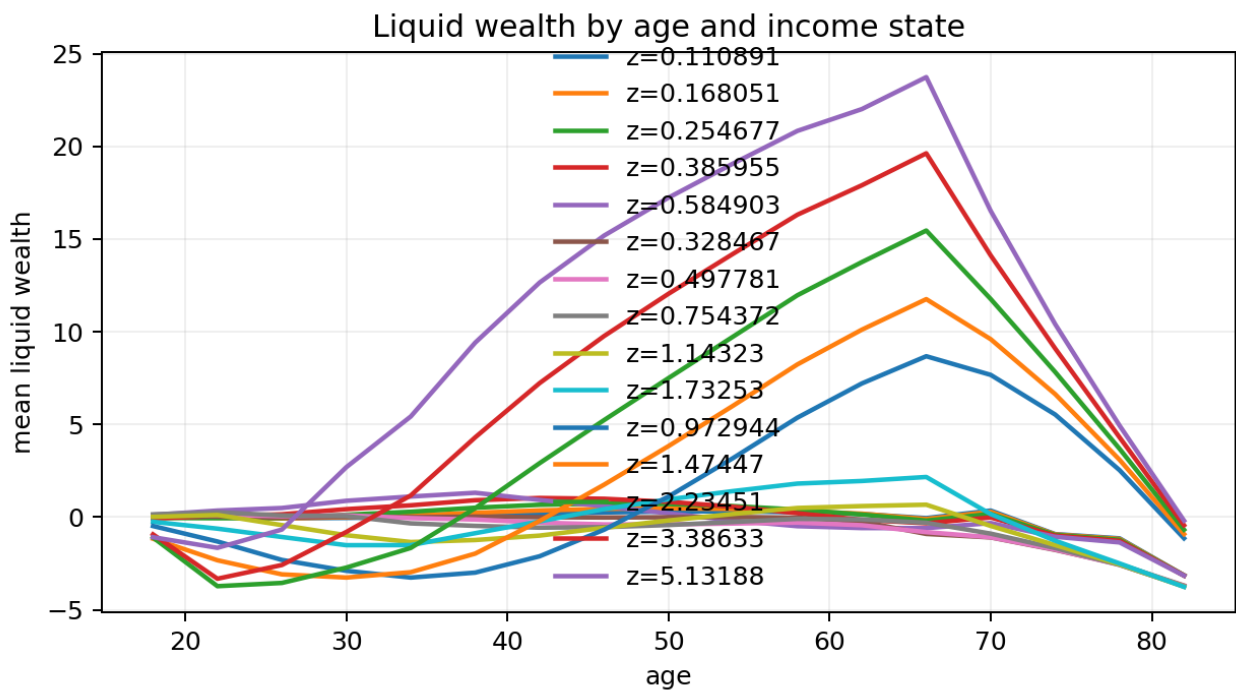
Baseline policy — ownership by age income state



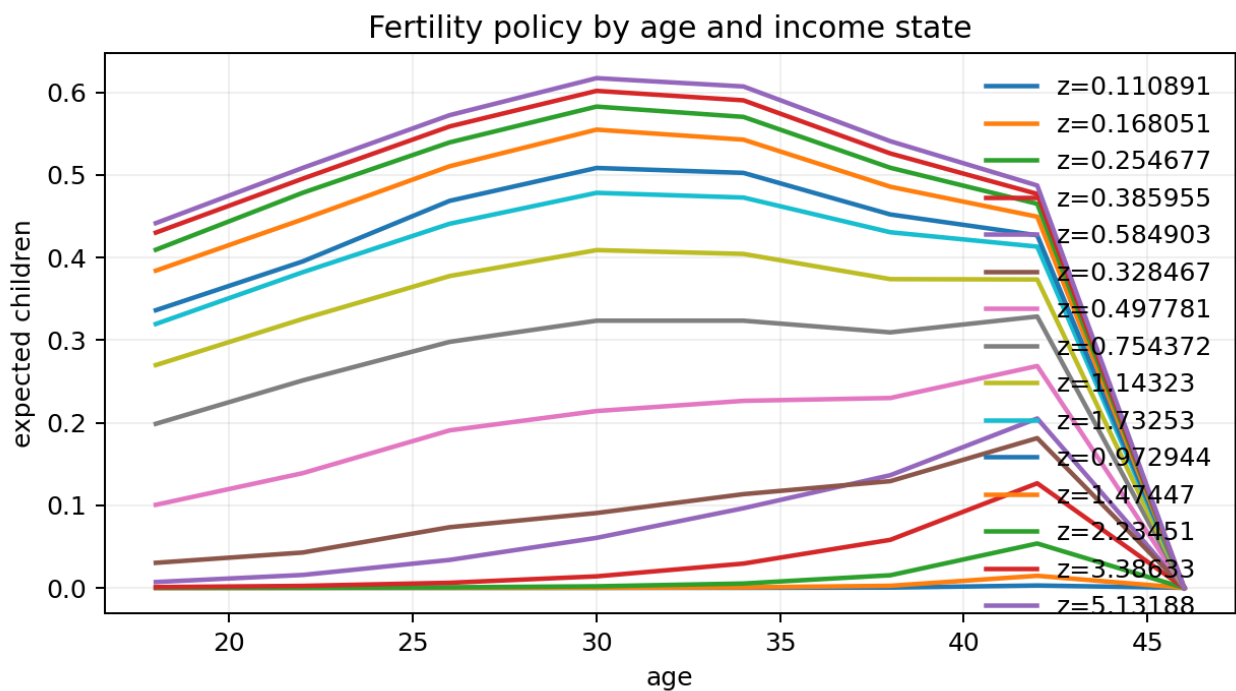
Baseline policy — housing by age income state

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



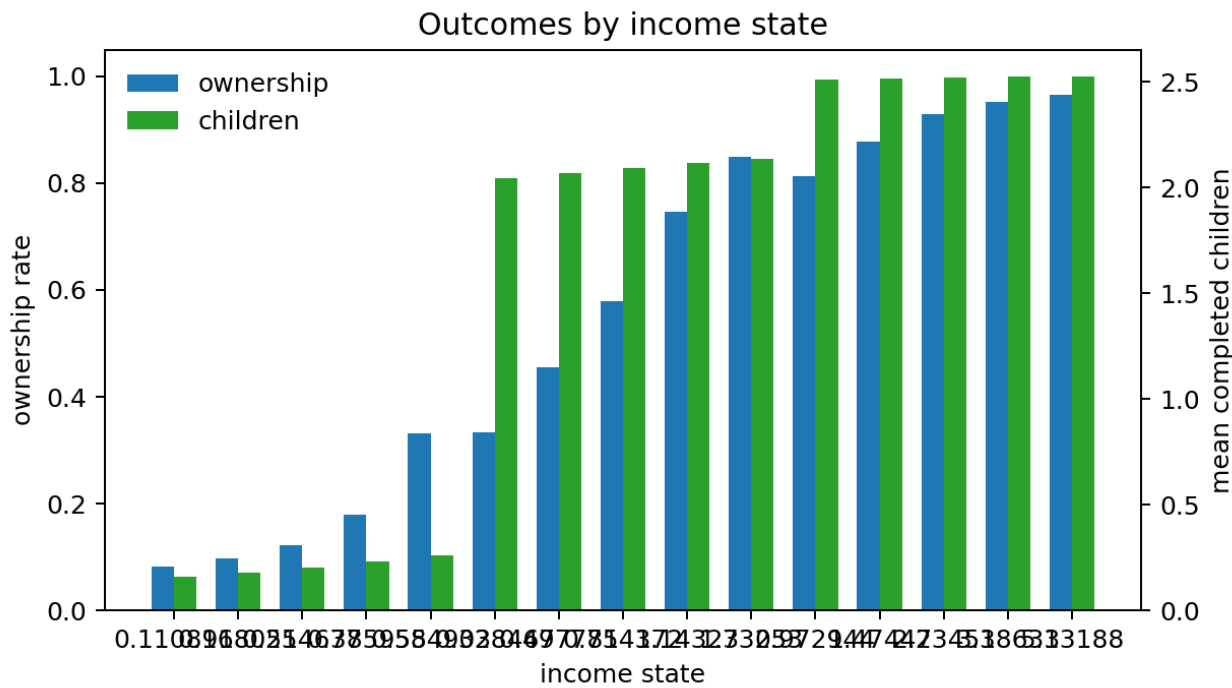
Baseline policy — liquid wealth by age income state



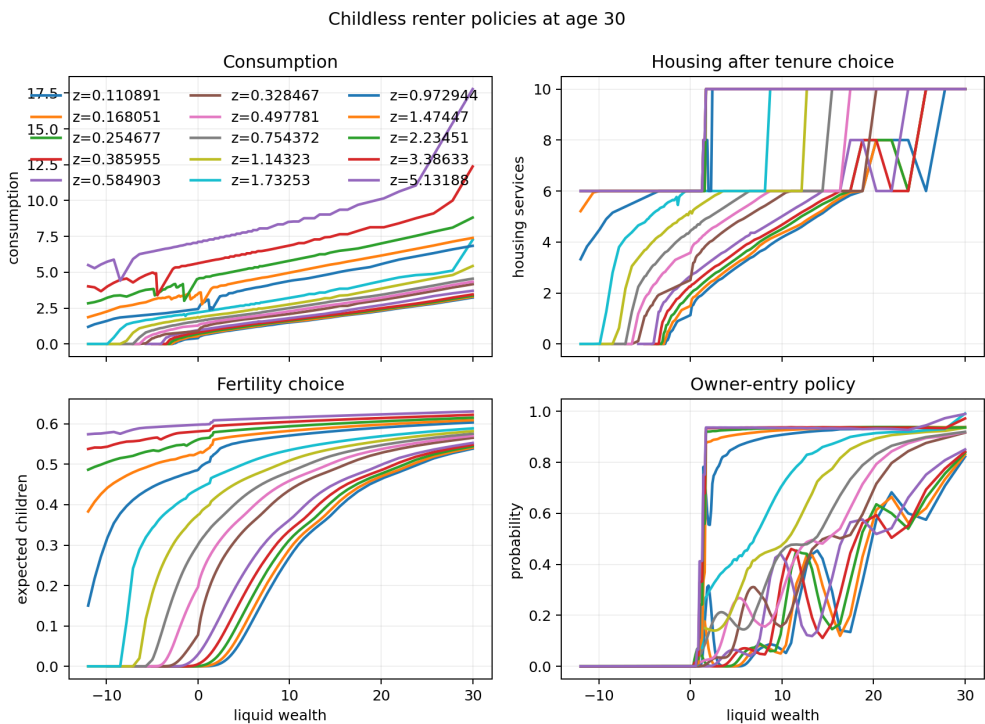
Baseline policy — fertility policy by age income state

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



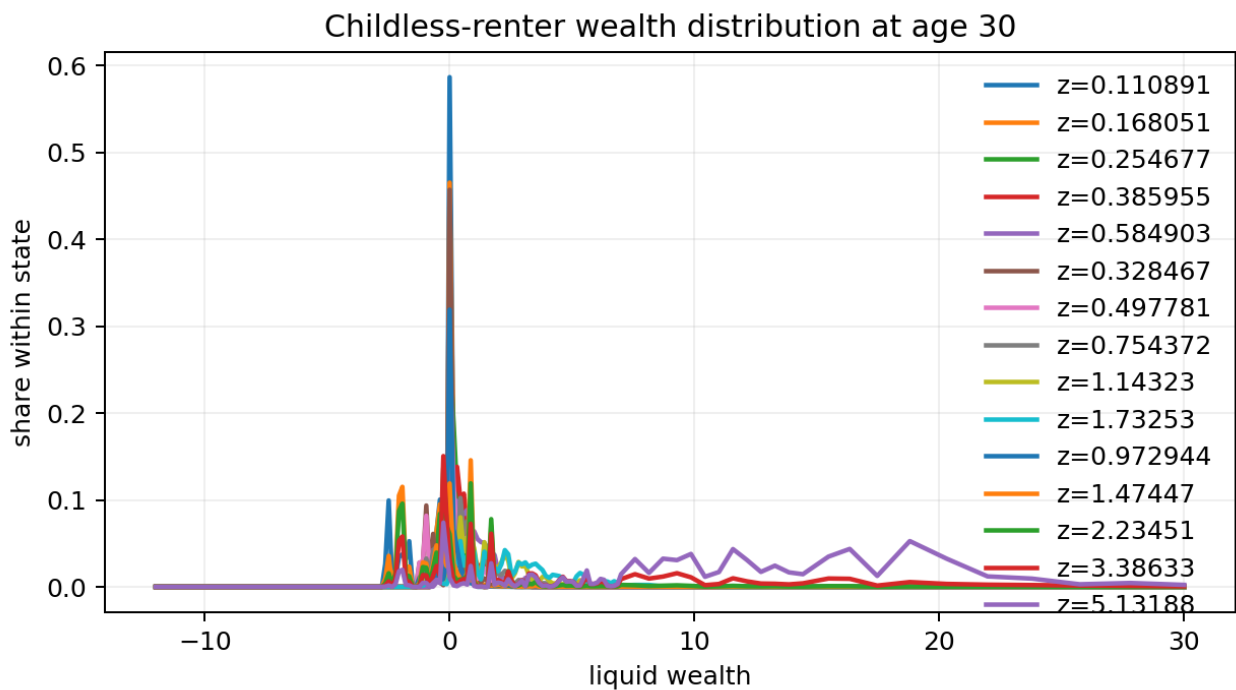
Baseline policy — income state outcomes



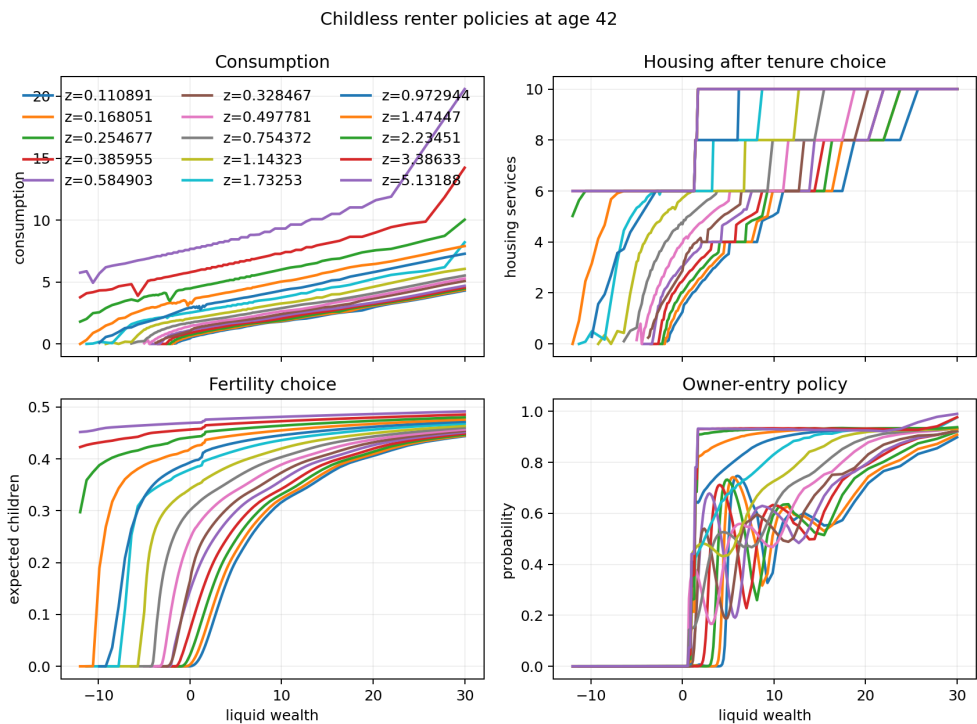
Baseline policy — policy childless renter age30

Policy native diagnostics: Baseline policy

The unchanged 17-figure standard diagnostic packet for this policy.



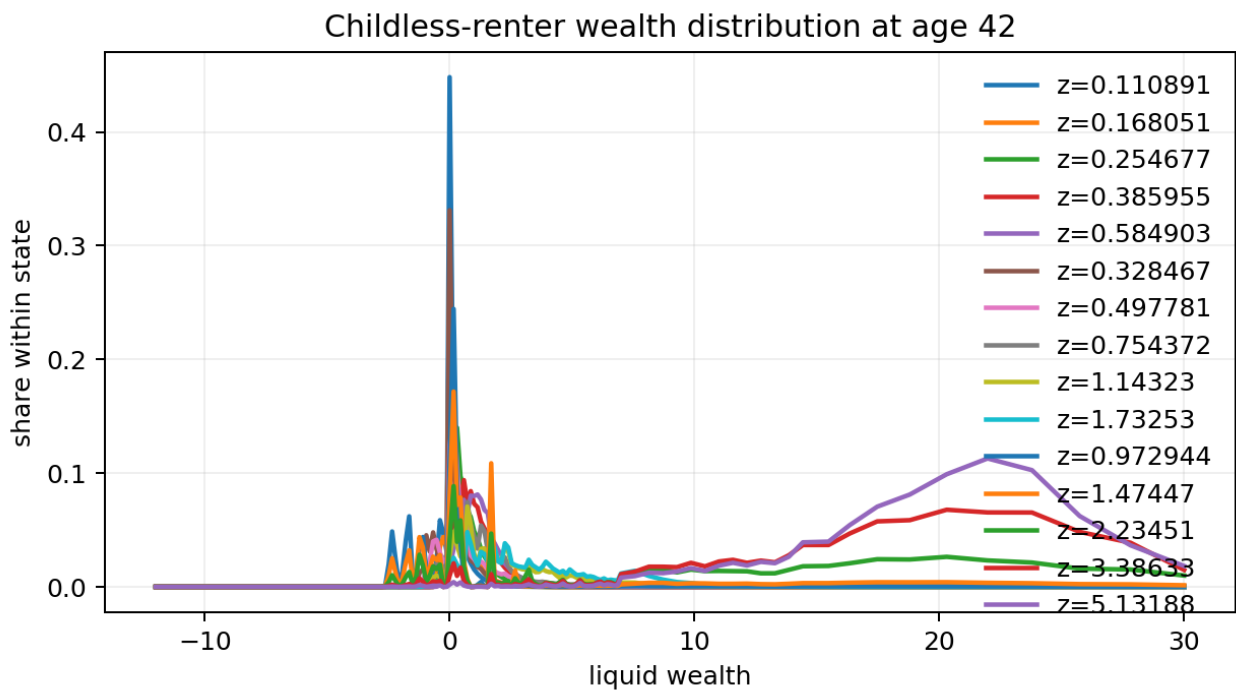
Baseline policy — wealth dist childless renter age30



Baseline policy — policy childless renter age42

Policy native diagnostics: Baseline policy

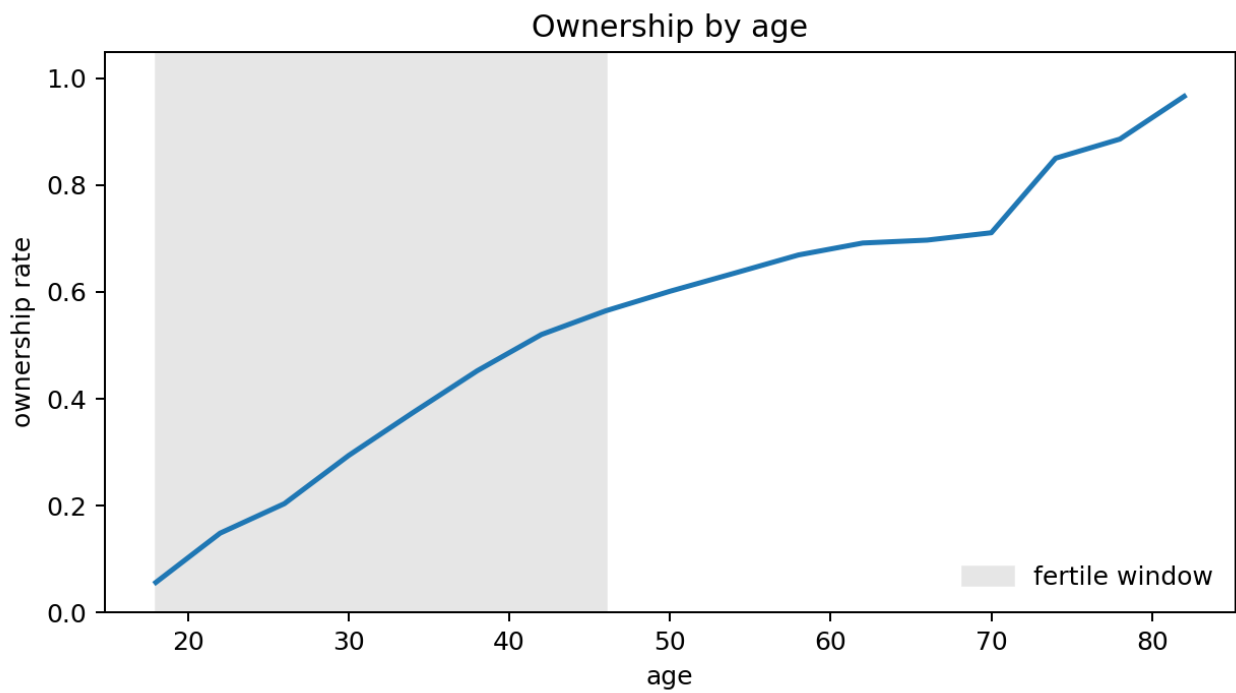
The unchanged 17-figure standard diagnostic packet for this policy.



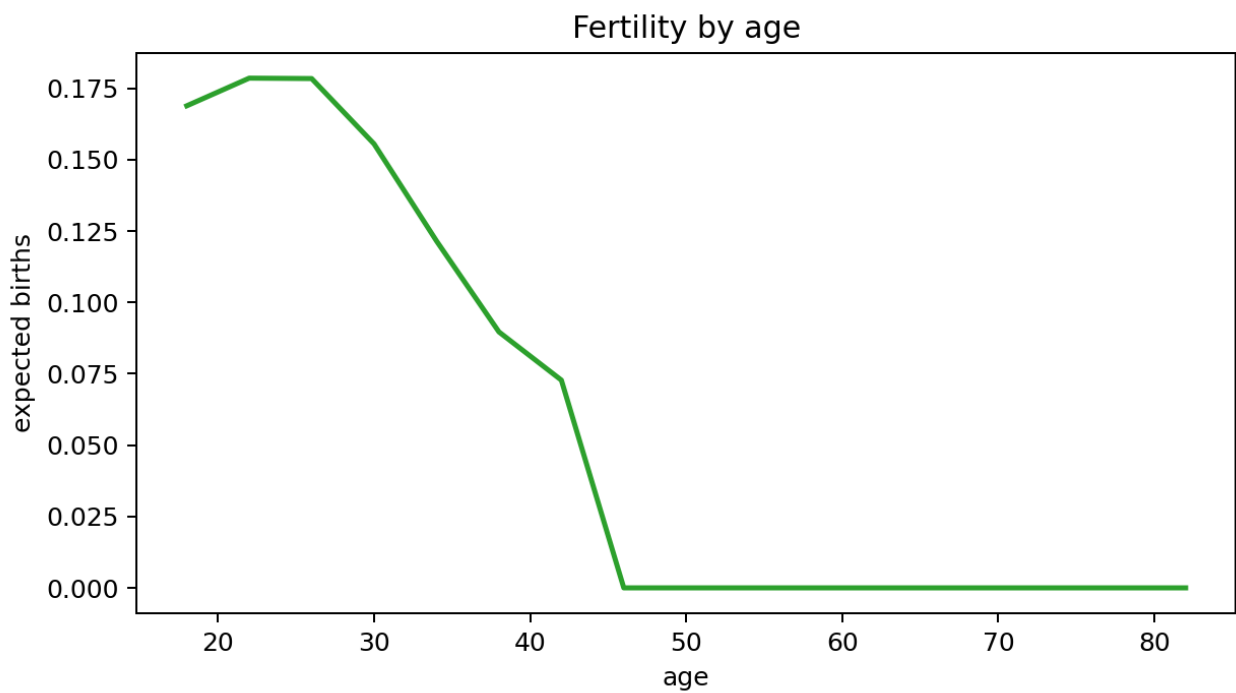
Baseline policy — wealth dist childless renter age42

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



2% tax policy — ownership by age



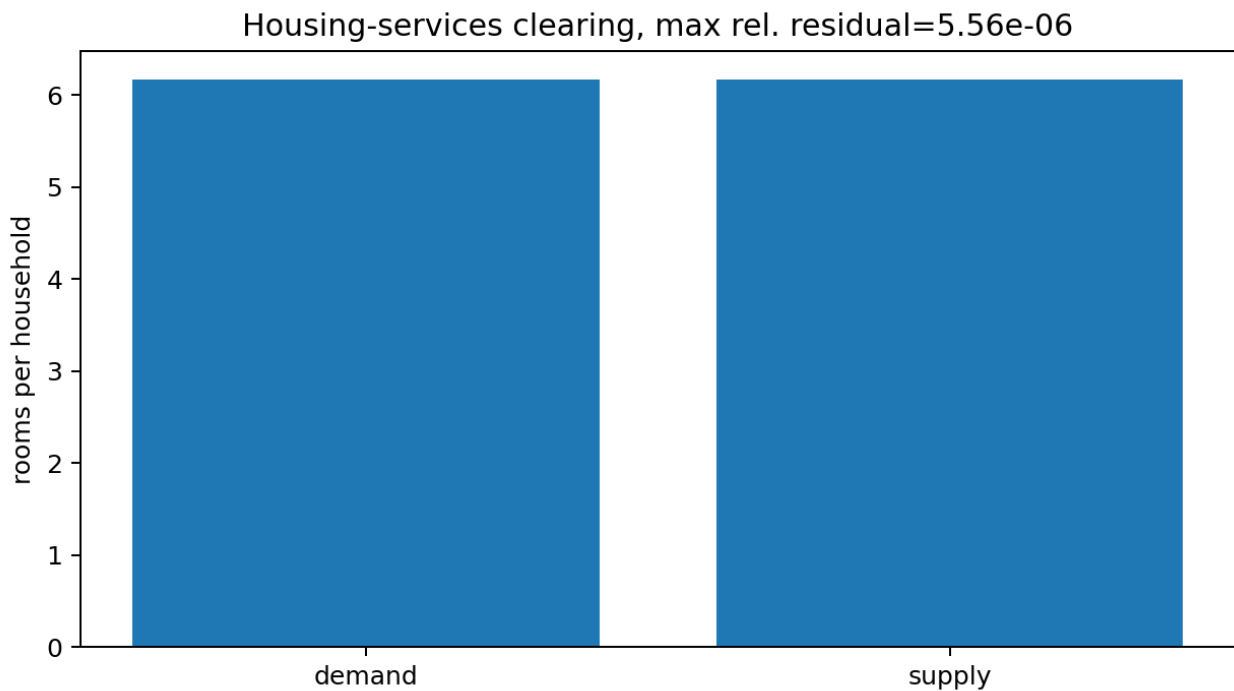
2% tax policy — fertility by age

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



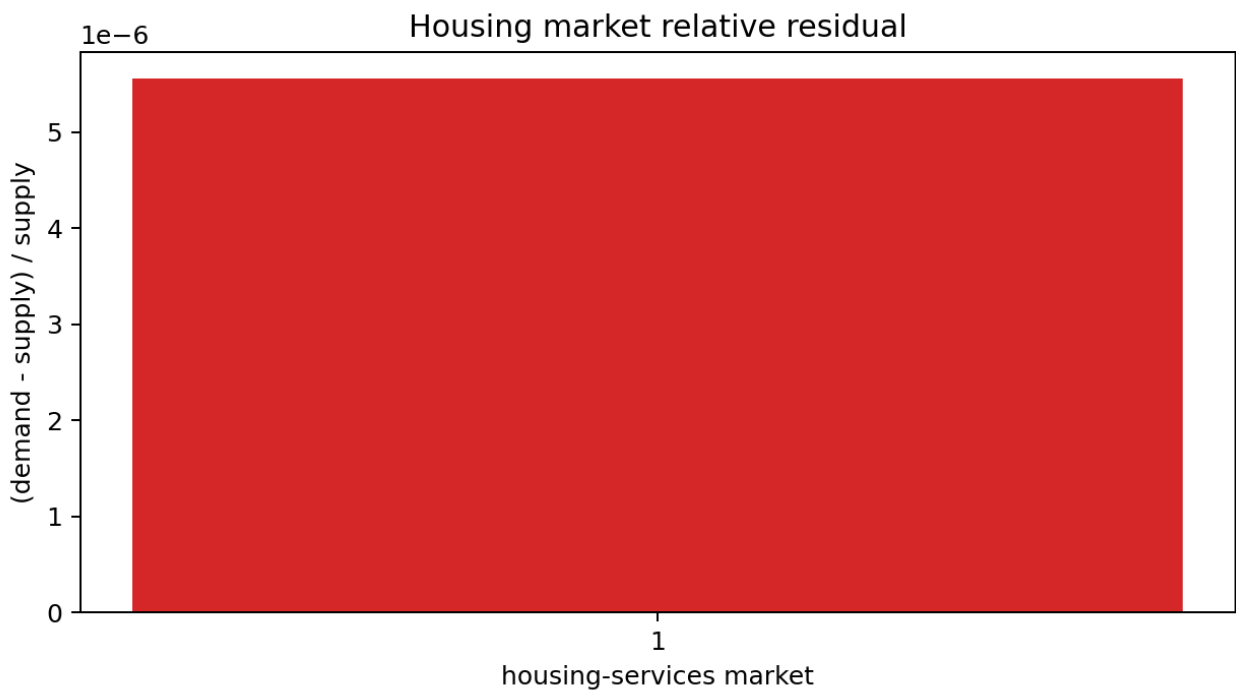
2% tax policy — housing prices



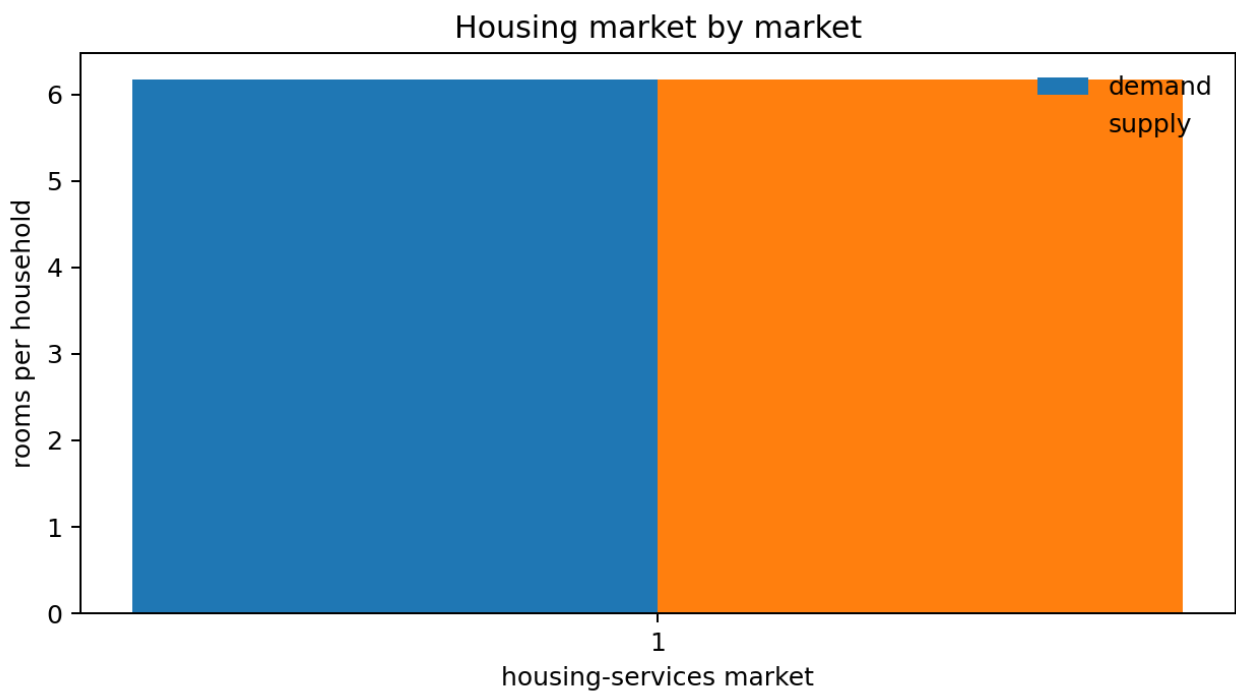
2% tax policy — housing market

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



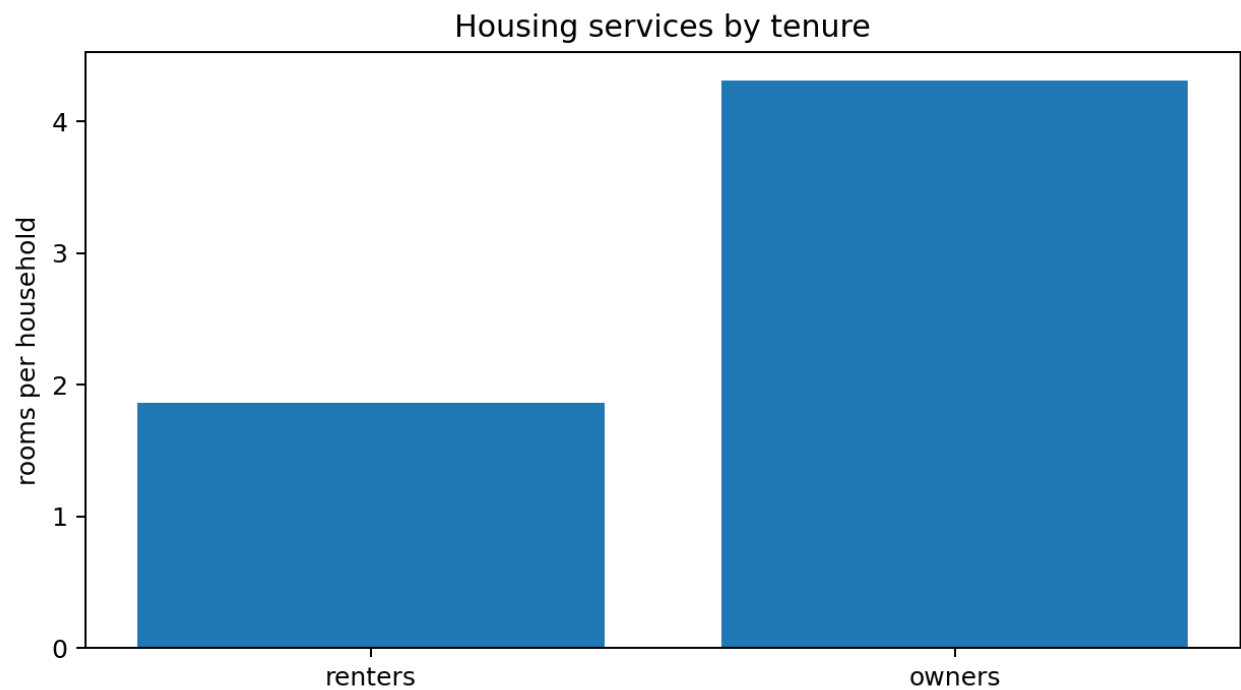
2% tax policy — market clearing residuals



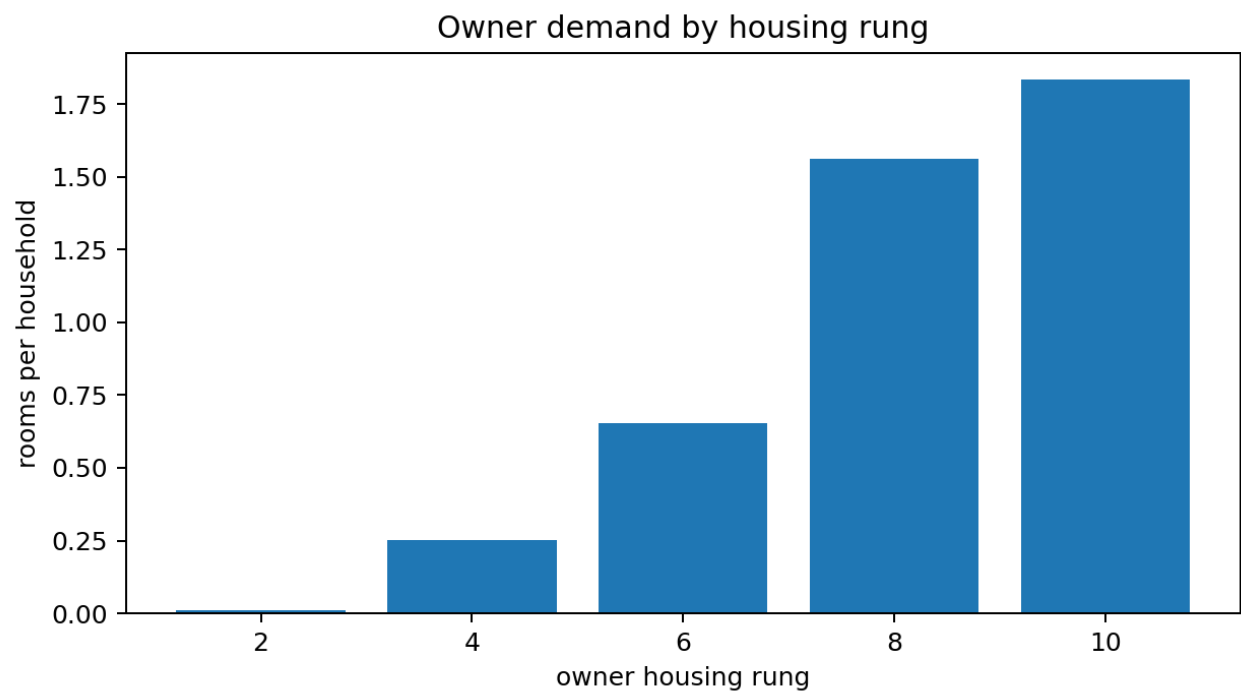
2% tax policy — market clearing by market

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



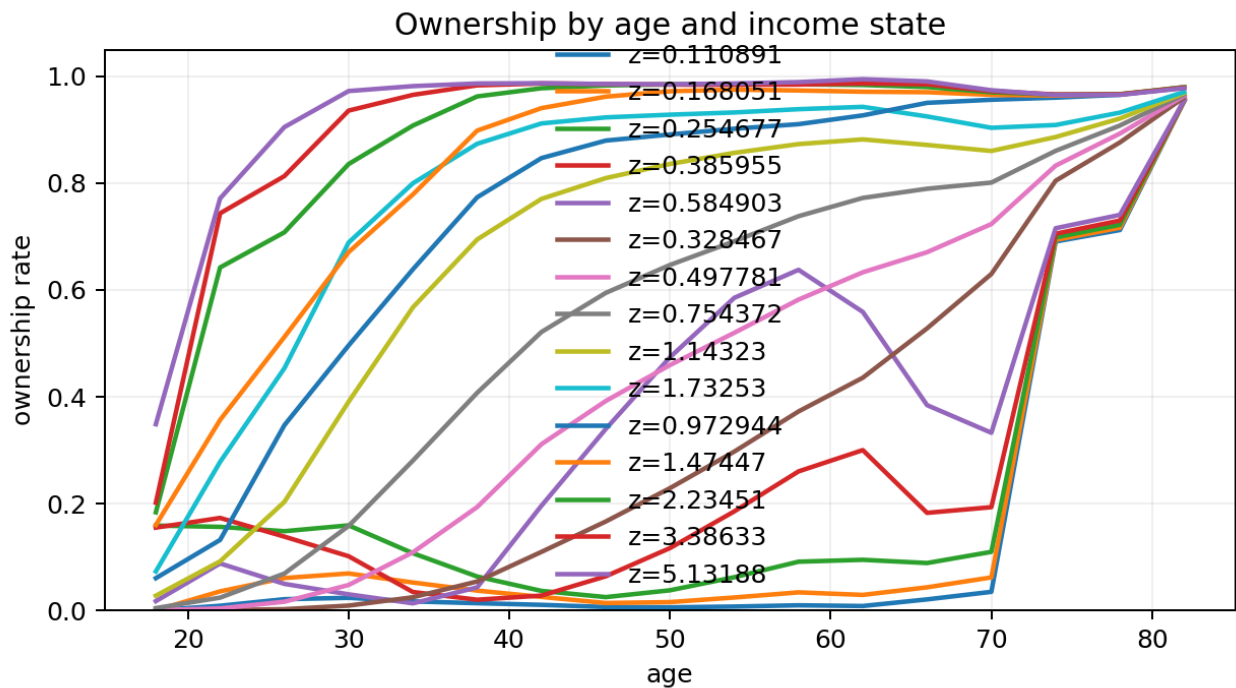
2% tax policy — tenure services



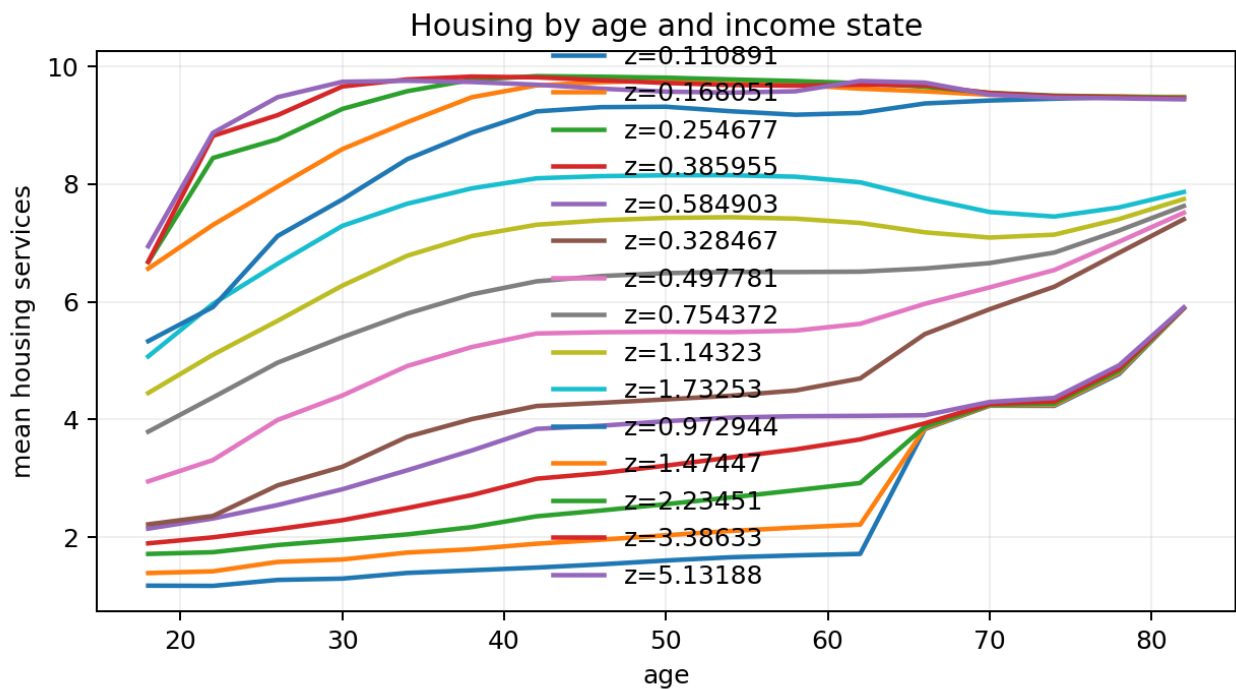
2% tax policy — owner rungs

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



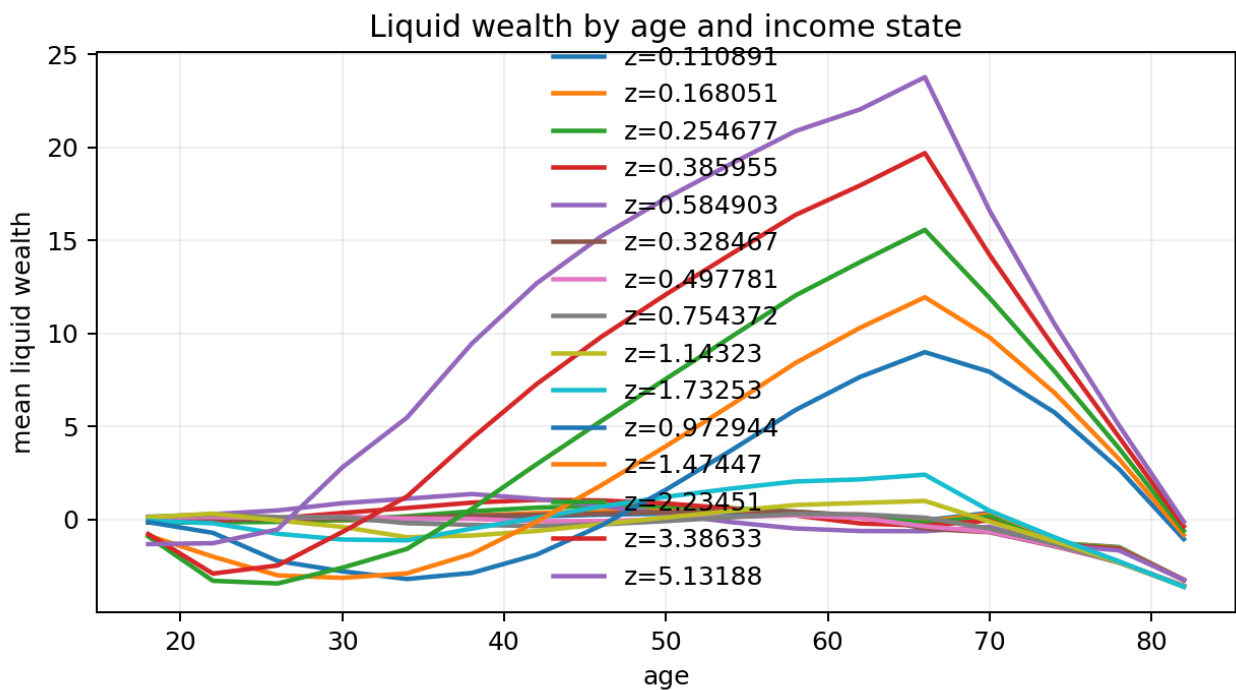
2% tax policy — ownership by age income state



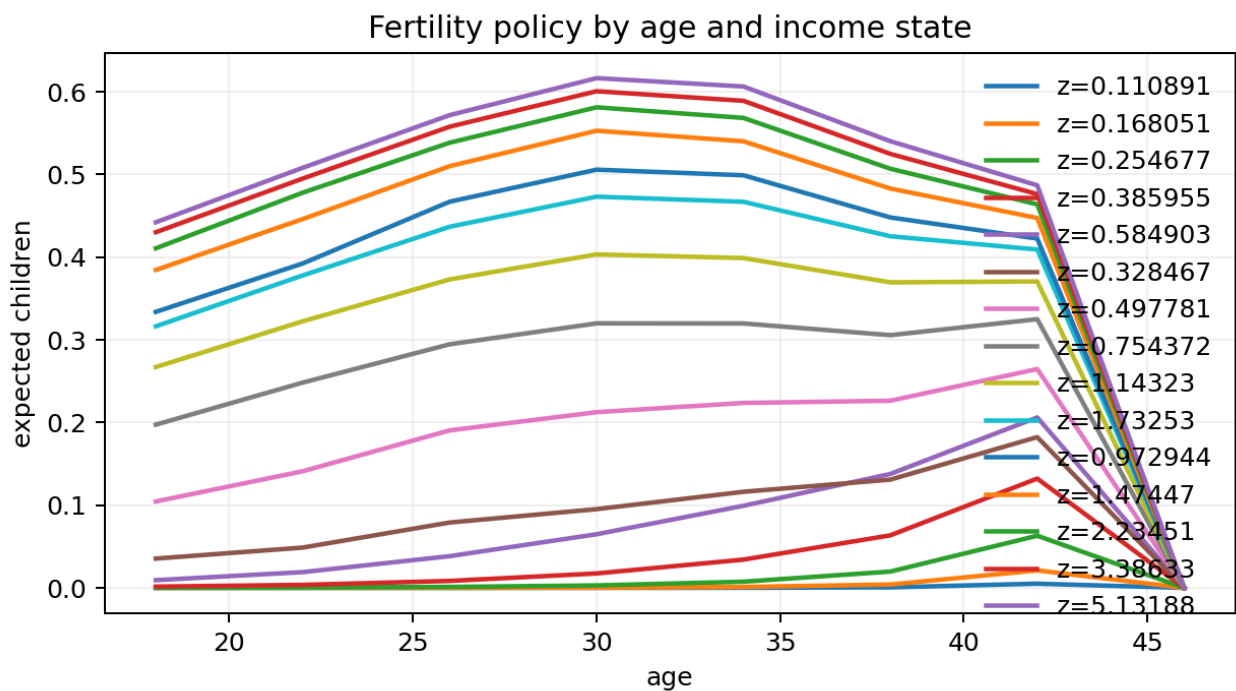
2% tax policy — housing by age income state

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



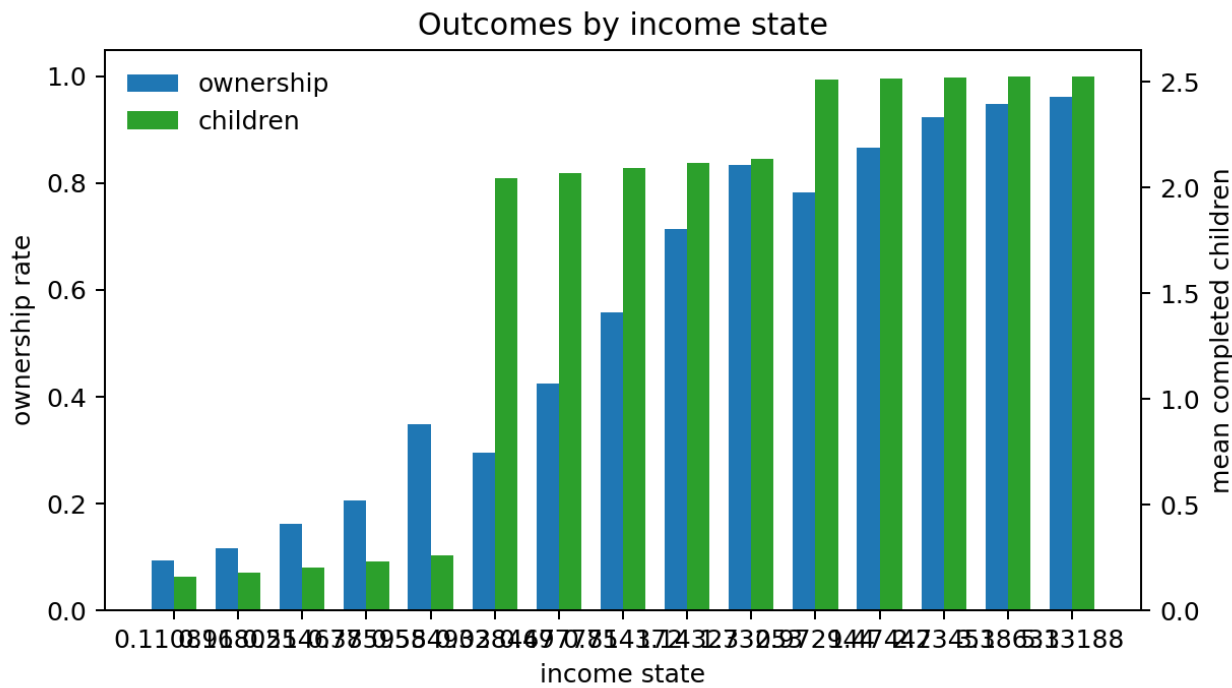
2% tax policy — liquid wealth by age income state



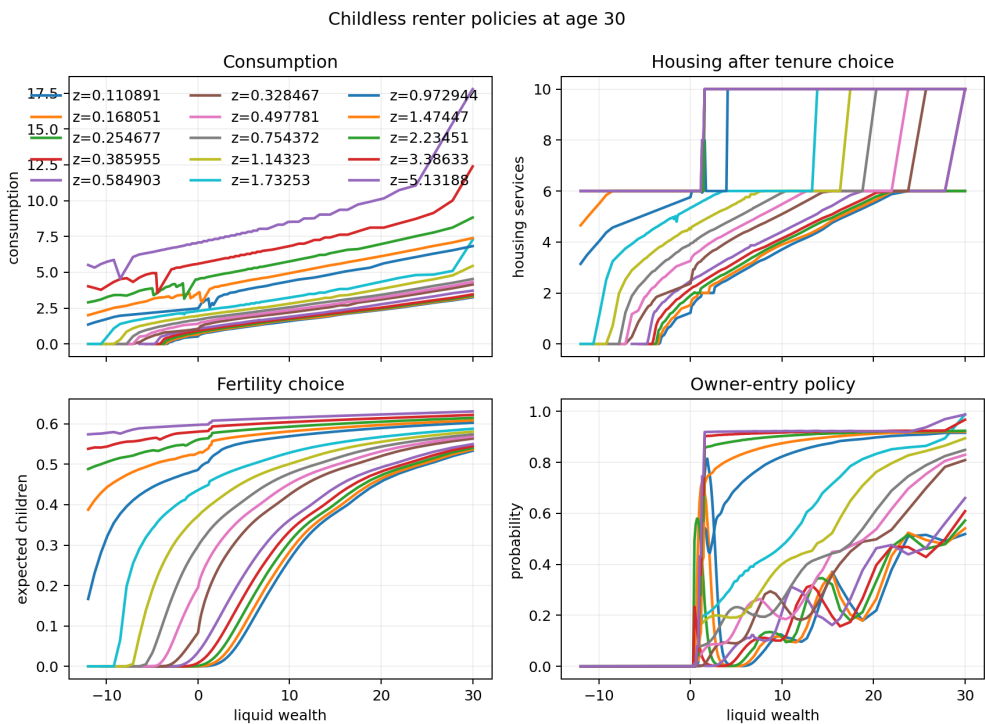
2% tax policy — fertility policy by age income state

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



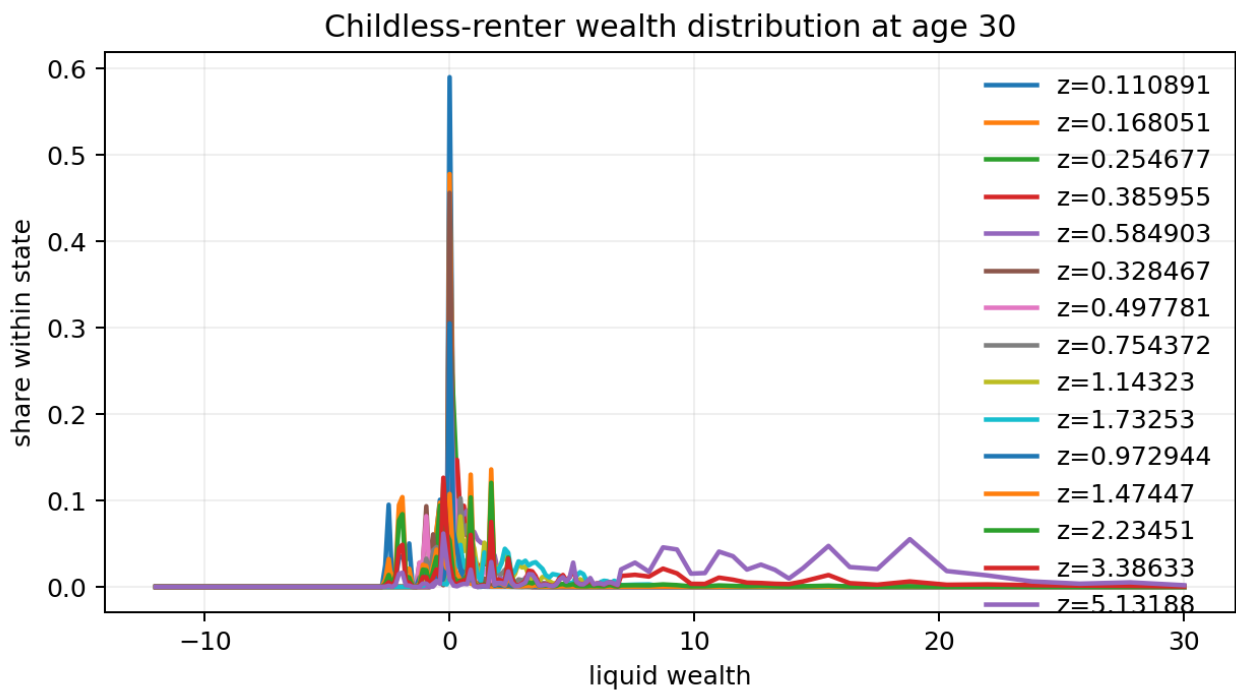
2% tax policy — income state outcomes



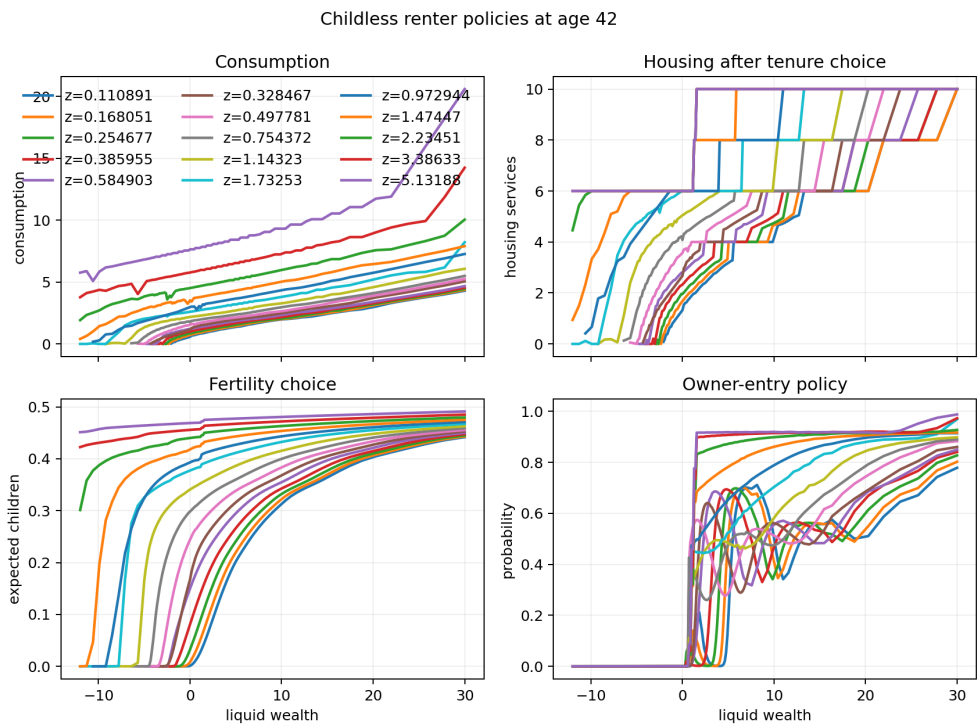
2% tax policy — policy childless renter age30

Policy native diagnostics: 2% tax policy

The unchanged 17-figure standard diagnostic packet for this policy.



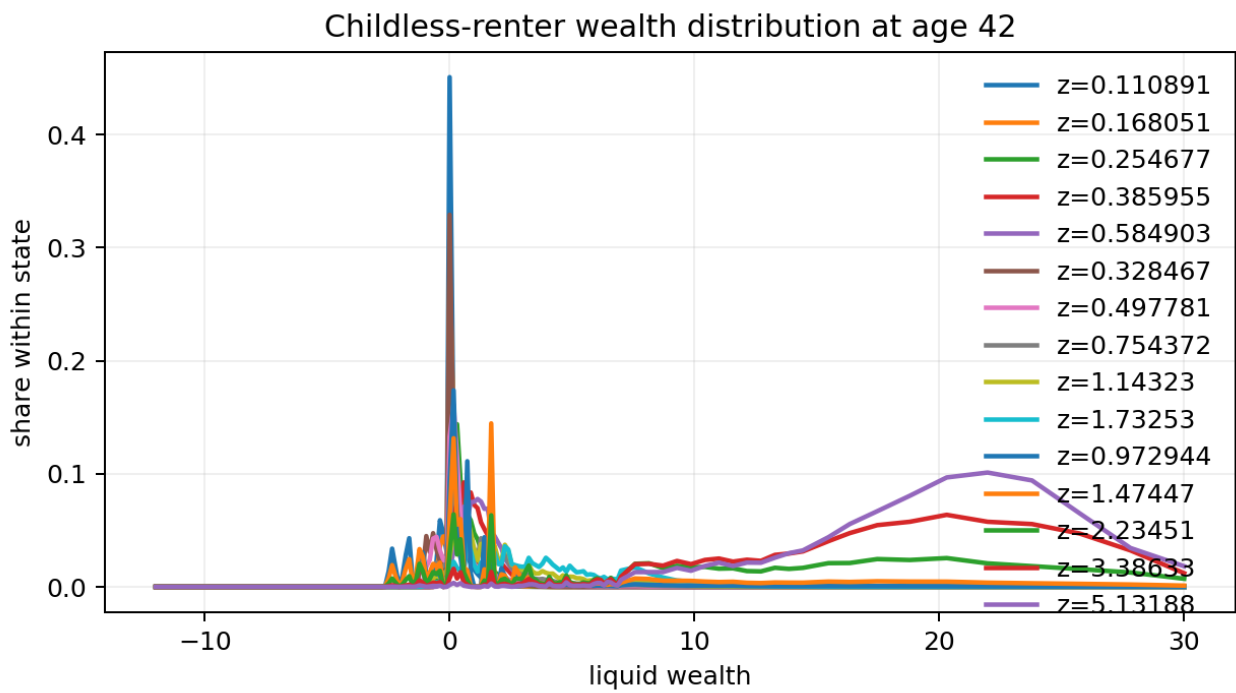
2% tax policy — wealth dist childless renter age30



2% tax policy — policy childless renter age42

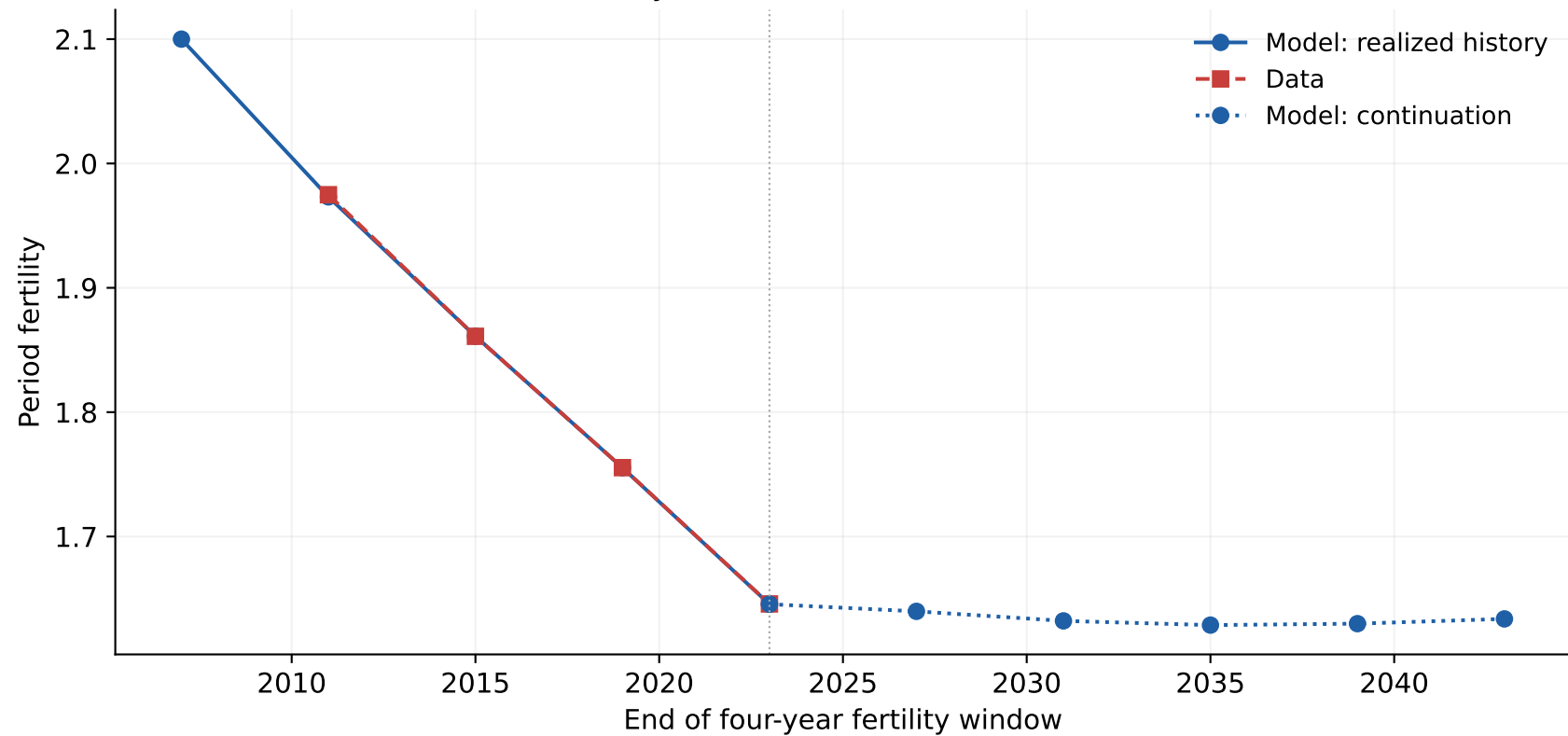
Policy native diagnostics: 2% tax policy

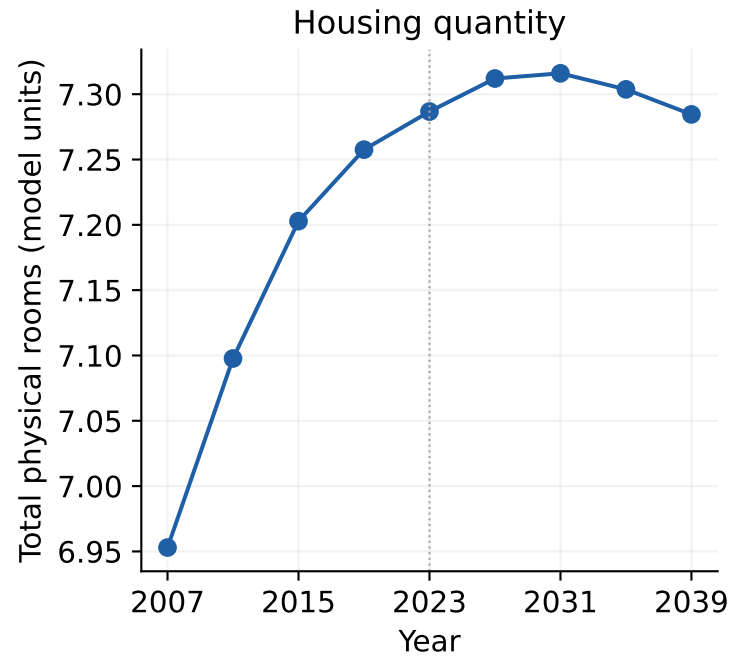
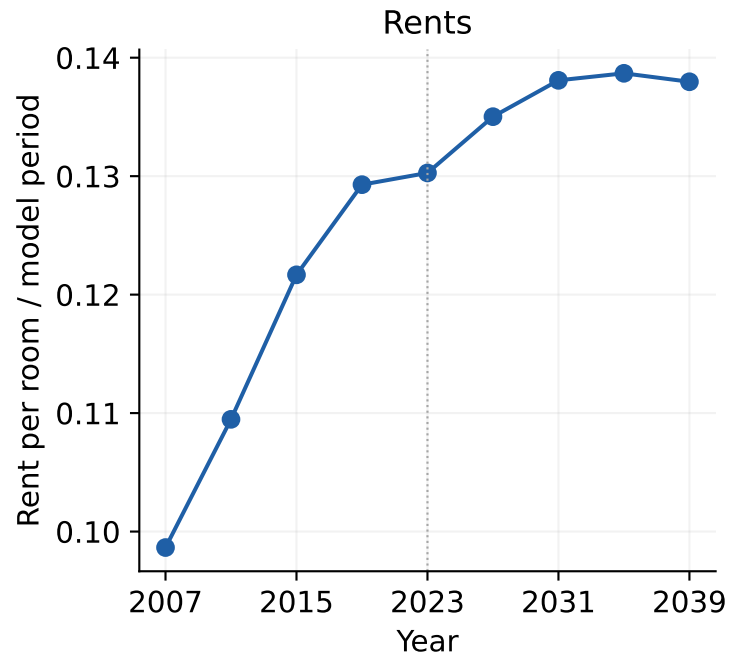
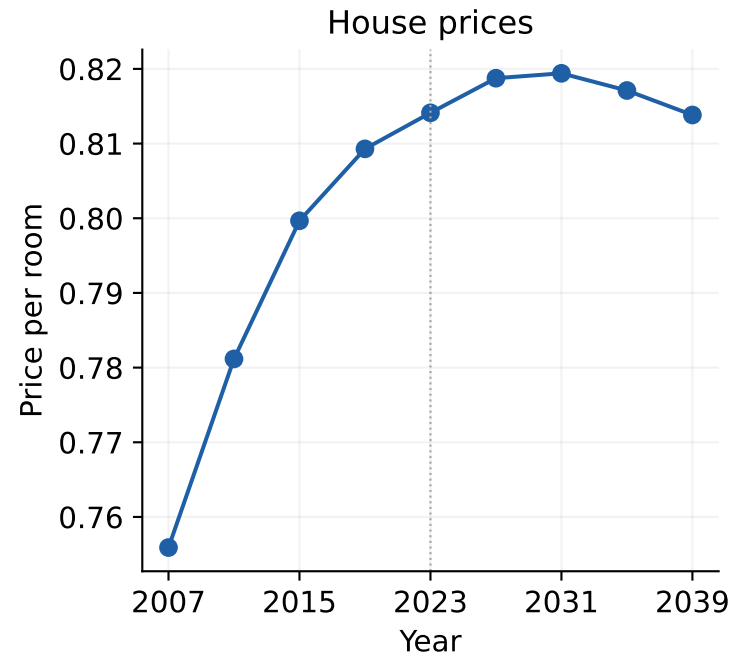
The unchanged 17-figure standard diagnostic packet for this policy.



2% tax policy — wealth dist childless renter age42

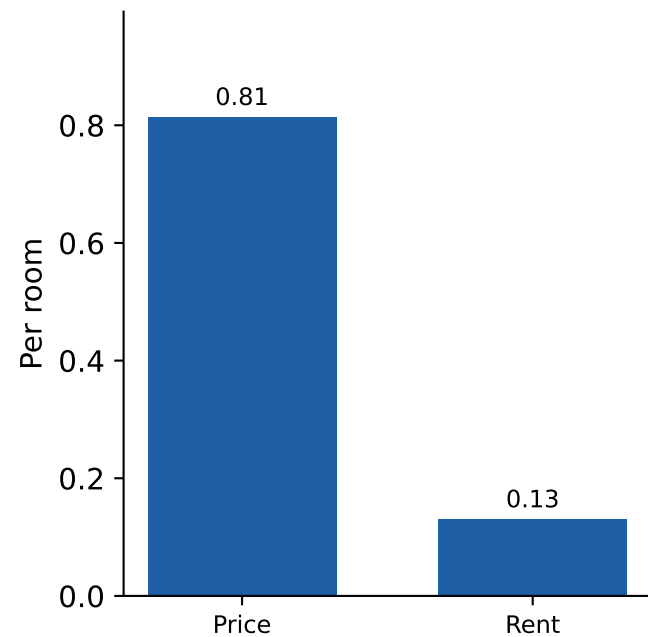
Fertility: model, data and continuation



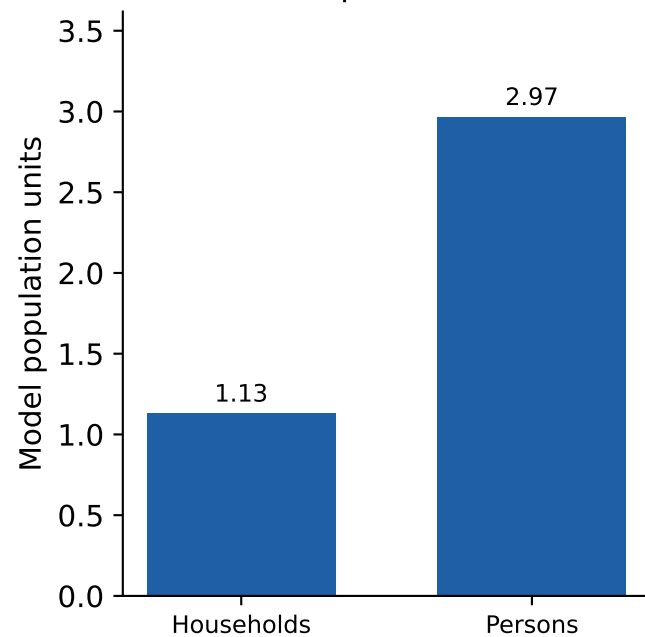


Finite horizon unverified; baseline property tax equally rebated; PAYGO balanced.

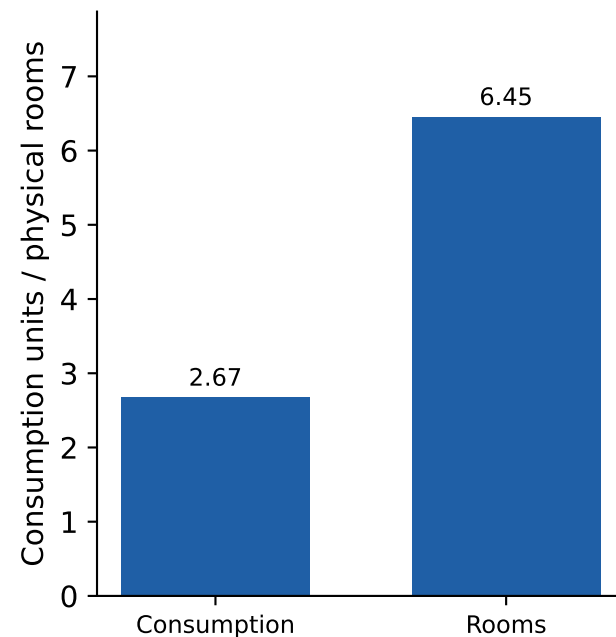
Prices



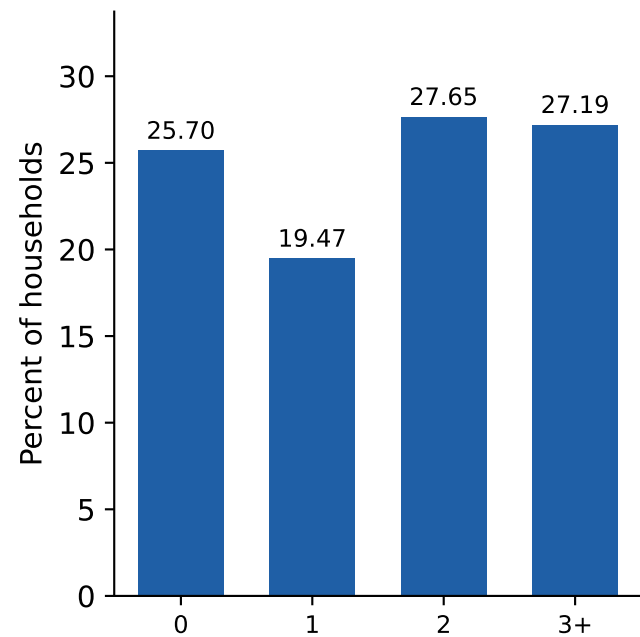
Population

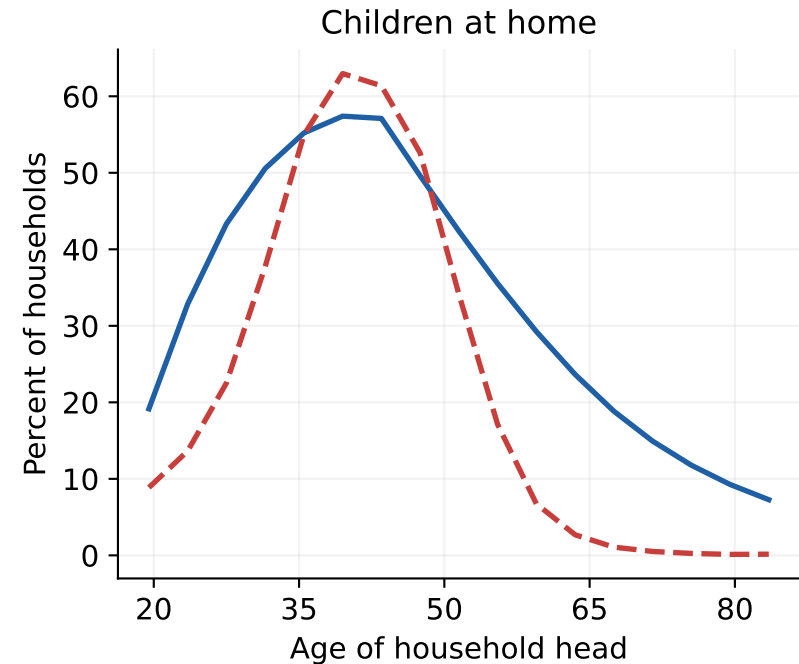
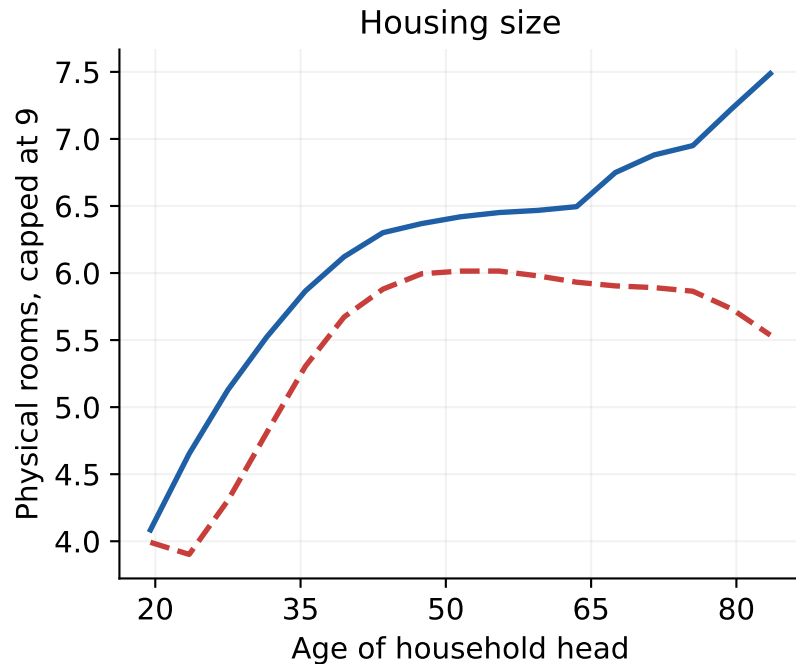
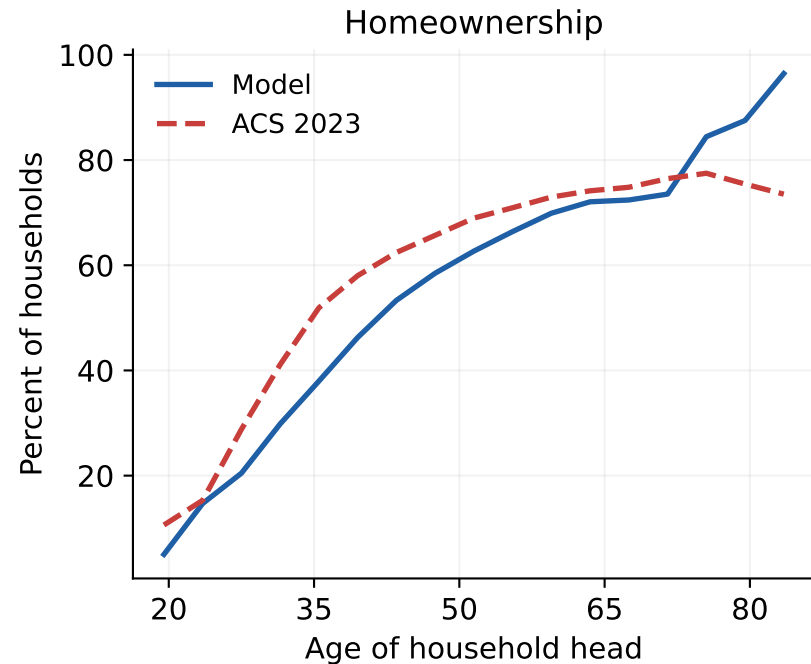


Household means



Children ever born

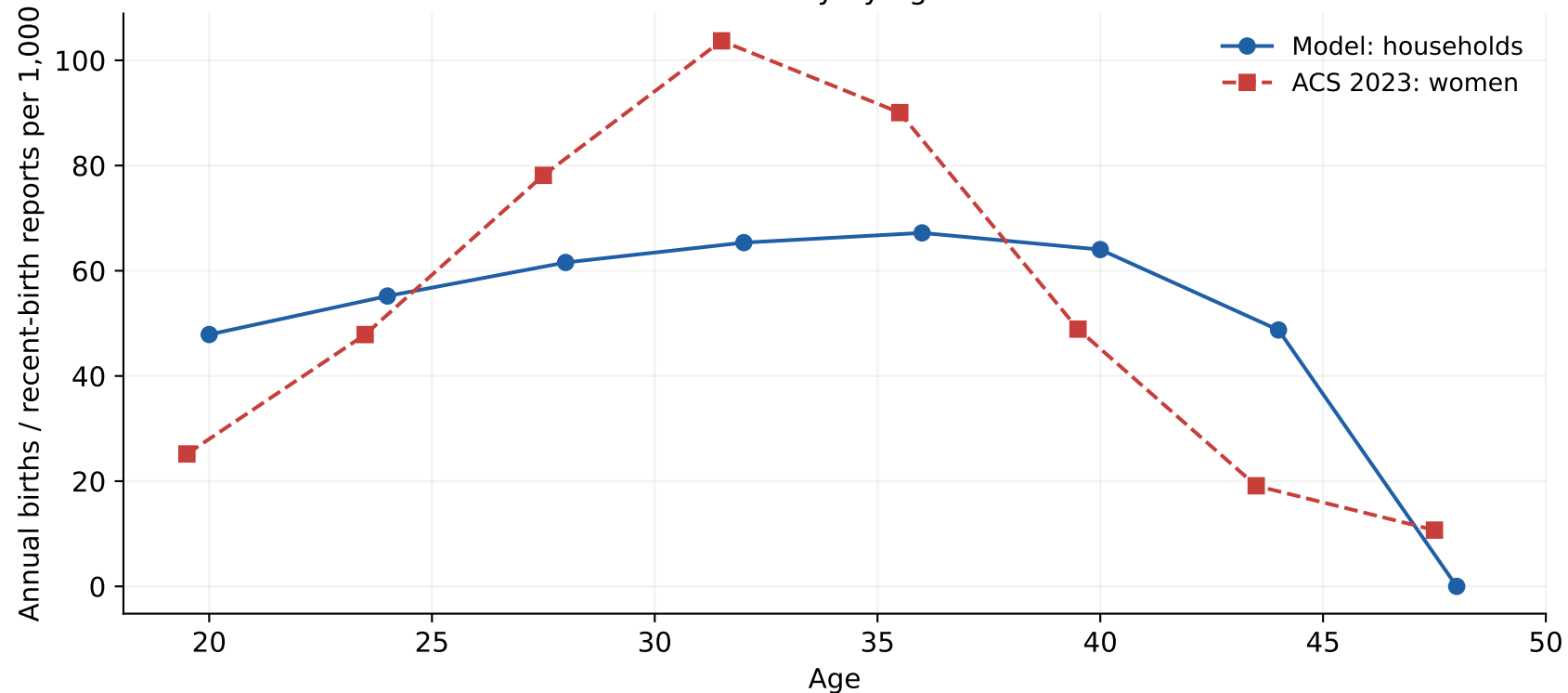




Children at home: model dependents; ACS resident own children under 18.

Finite horizon unverified; baseline property tax equally rebated; PAYGO balanced.

Fertility by age

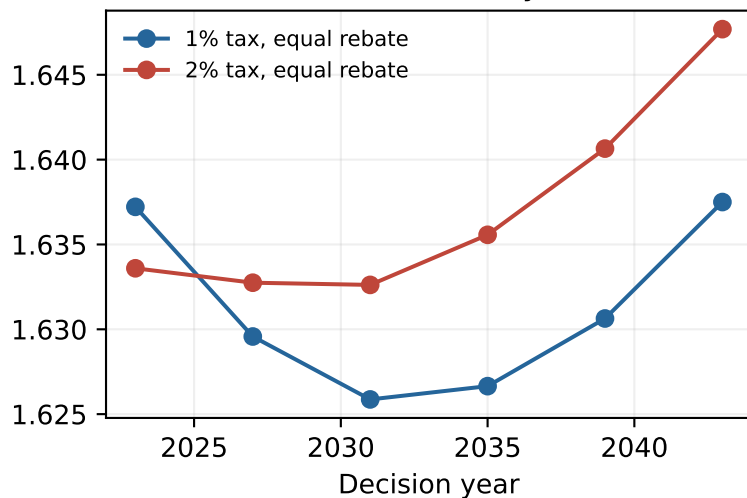


Model: annualized 2024–2027 birth flow. ACS: women reporting a birth in the previous 12 months.

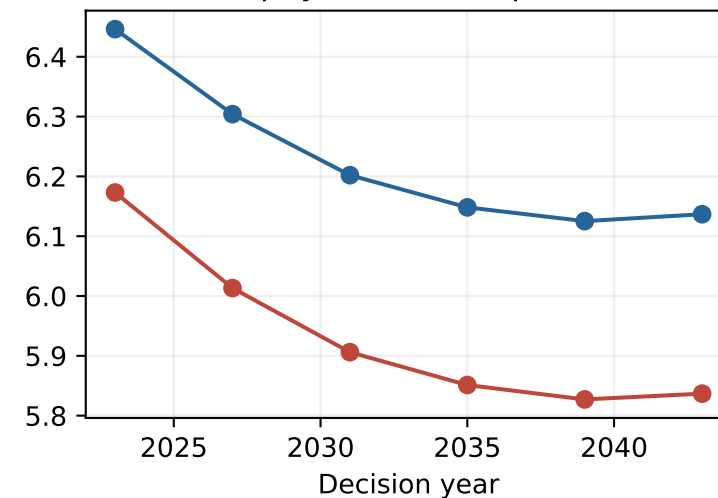
Finite horizon unverified; baseline property tax equally rebated; PAYGO balanced.

Property tax with equal rebates and balanced pensions

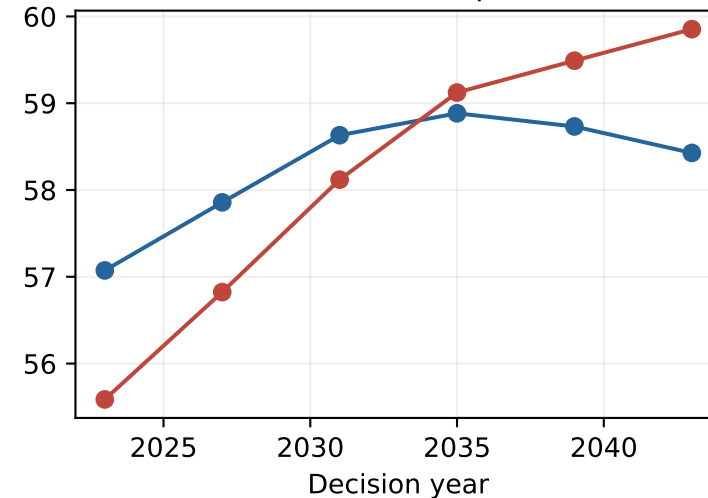
Period fertility



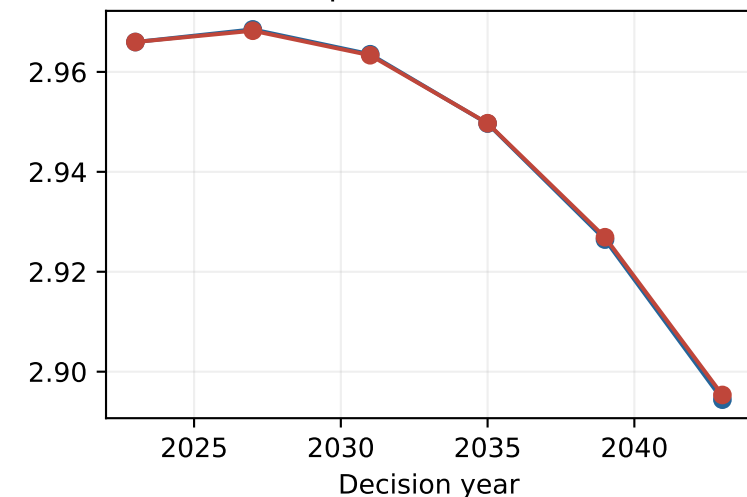
Mean physical rooms per head



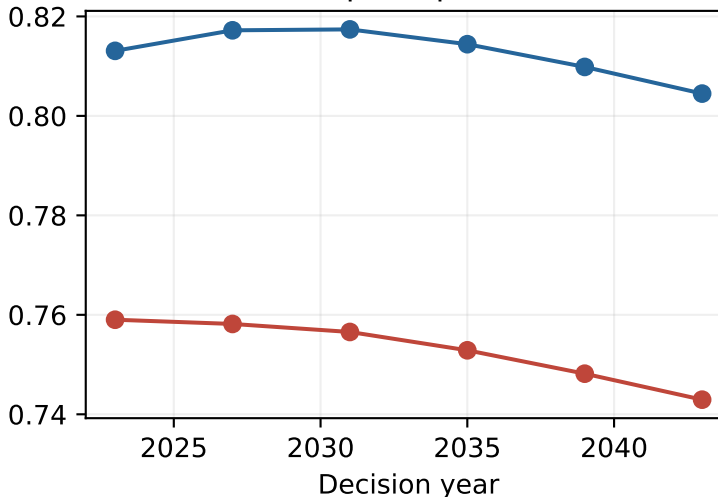
Homeownership (%)



Resident persons (model units)



House price per room



Pension per model period

